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Supplementary information

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Six-state clock physics in an atomically thin antiferromagnet

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Supplementary Note 1: Raman signatures of magnetic ordering

In this section, we explore the Raman signatures of magnetic ordering in $NiPS_3$, with a particular focus on the monolayer limit. Our analysis centers on two key features: the bimagnon continuum, which reflects the presence of short-range antiferromagnetic correlations, and the P_2 phonon mode, which signals the breaking of rotational symmetry and has been associated with the emergence of spin order[1,2].

Bimagnon continuum: We begin by examining the bimagnon continuum in few-layer $NiPS_3$ to gain insight into its magnetic ordering and phase transitions. We collect temperature-dependent Raman spectra from samples ranging from monolayer (1L) to tetralayer (4L) in the cross-circularly polarized $\sigma^+\sigma^-$ channel, with the results presented in Fig. S1. In these spectra, the sharp peak at 65 meV (white peak) corresponds to a phonon mode in the silicon substrate, while the broad feature centered around ∼68 meV (yellow feature) is the bimagnon scattering continuum, associated with double spin-flip processes mediated by exchange interactions.

This bimagnon continuum is observed down to the monolayer, consistent with previous studies[3], and indicates the presence of short-range magnetic correlations in all thicknesses. From the color maps, we observe that both the peak position (ω_0) and linewidth of the bimagnon continuum exhibit abrupt changes at T $\simeq$ 150 K across all layer thicknesses. To quantify this behavior, we extract the fitted peak position ω_0 and its temperature derivative, $d\omega_0/dT$ and plot them in Fig. S2a-b, respectively.

Notably, ω_0 shows an inflection point, while $d\omega_0/dT$ displays a peak near $T \sim$ 150-160 K, in agreement with previous studies[3]. These results suggest the onset of magnetic correlations around this temperature range and are consistent with the possible occurrence of a magnetic phase transition. However, this measurement alone does not provide direct access to the critical scaling behavior of the magnetic order.

P_2 phonon mode: The complete raw Raman scattering data at lower Stokes shifts are presented in Fig. S3 for samples of varying thickness. A signature of magnetic ordering is observed in the splitting of the P_2 phonon peak at ∼22 meV. As shown in Figs. 2d,f, this splitting (ΔP_2) is unambiguously detected as a function of temperature in thin flakes down to the bilayer. In the monolayer, a small but discernible ΔP_2 persists up to ∼120 K (Fig. S4a,b), suggesting the presence of magnetic correlations. Notably, this temperature coincides with the Néel temperature (T_N) determined independently from the second harmonic generation (SHG) anisotropy and intensity signals (Fig. S4c,d). This behavior can be understood in the context of the phonon mode splitting (ΔP_2) observed in thicker samples. Below T_N, the magnetic order in monolayer $NiPS_3$ couples with the E_g phonon mode, lifting its degeneracy and forming a pair of A_g and B_g modes. However, as long-range order is lost and the Berezinskii–Kosterlitz–Thouless (BKT) phase appears between $T_N < T < T_{BKT}$, the E_g mode no longer splits. In the paramagnetic phase above T_{BKT}, the E_g mode remains doubly degenerate. Early studies may have also observed a small but finite splitting in monolayer $NiPS_3$[1], although in that case, the splitting was considered within experimental error, and the null hypothesis was assumed. It was argued that the disappearance of ΔP_2 in monolayer $NiPS_3$ indicated the complete suppression of magnetic ordering. However, a follow-up study suggested that the suppression of ΔP_2 might also be related to the increased crystal symmetry in the monolayer system[4]. More recently, a splitting similar to ours was reported at 4 K; however, no systematic investigation of its temperature evolution was conducted[5]. Our results are broadly consistent with these recent findings: the weak splitting of P_2 observed in the monolayer is consistent with our SHG measurements and points to the onset of magnetic ordering below $T_N \sim 120$ K.

Supplementary Note 2: Spin Hamiltonian and magnetocrystalline anisotropy

In this section, we perform a group theory analysis of multilayer and monolayer $NiPS_3$ to motivate the spin Hamiltonian and magnetocrystalline anisotropies used in this work.

The point group of multilayer/bulk $NiPS_3$ is monoclinic $2/m$, which yields, to the lowest order, a magnetocrystalline anisotropy energy of

$$E_a = K_2' \sin^2\theta + K_2'' \sin^2\theta \cos(2\psi), \tag{S1}$$

where K_2' and K_2'' are the out-of-plane and in-plane anisotropy components, respectively, θ is the polar angle measured from the out-of-plane c^*-axis, and ψ is the in-plane azimuthal angle measured from the crystallographic a axis.[6] Neutron diffraction experiments on the bulk crystal indicate that $K_2' < 0$ and $K_2'' < 0$, with $|K_2'| \gg |K_2''|$, favoring a predominantly in-plane spin alignment with a weak magnetic easy axis along a.[7] This leads to an in-plane $\mathbb{Z}_2$ symmetry-breaking spin order, placing the system within the 3D Ising universality class. However, in the few-layer limit where the monoclinic distortion weakens, the anisotropy becomes less pronounced, and the system is expected to crossover toward a 3D XXZ Heisenberg behavior.

In contrast to the bulk, monolayer $NiPS_3$ crystallizes in a trigonal structure with point group $\bar{3}m$, which lifts the in-plane $\mathbb{Z}_2$ anisotropy characteristic of the monoclinic structure. The lowest-order symmetry-allowed magnetocrystalline anisotropy energy in this case takes the form[6]

$$\begin{aligned} E_a = K_2' \sin^2\theta + K_4' \sin^4\theta + K_4'' \sin^3\theta \cos\theta \cos(3\psi) + K_6' \sin^6\theta + \\ K_6'' \sin^6\theta \cos(6\psi) + K_6''' \sin^3\theta \cos^3\theta \cos(3\psi). \end{aligned} \tag{S2}$$

Since the magnetocrystalline anisotropy originates from a combination of spin-orbit and orbital-lattice interactions, and the van der Waals coupling between layers is weak, the monolayer is expected to retain the strong easy-plane anisotropy observed in the bulk. That is, the combination $K_2' + K_4' + K_6' \ll 0$ favors a spin orientation within the monolayer plane (*i.e.*, $\theta = \pi/2$). This constraint underpins a 2D XY magnetic order, modulated by the leading in-plane anisotropy term allowed by symmetry:

$$E_a = K_6'' \cos(6\psi) \equiv K_6 \cos(6\psi), \tag{S3}$$

which is the form of the magnetocrystalline anisotropy used in our Monte Carlo simulations. Furthermore, we note that the $\mathbb{Z}_3$ anisotropy terms (K_4'' and K_6''') are suppressed by the antiferromagnetic exchange interactions in the system. In a strong antiferromagnet, the magnetic anisotropy energy arising from the bipartite spin sublattices acquires opposite signs for any odd-order in-plane anisotropy terms. As a result, such contributions cancel out. Therefore, even under more relaxed assumptions about the strength of the out-of-plane anisotropy, the leading in-plane term in a trigonal antiferromagnet remains $\mathbb{Z}_6$.

Finally, we note that other sources of magnetic anisotropy beyond single-ion, including antisymmetric and anisotropic symmetric exchange interactions, are important for a variety of van der Waals magnets. However, the high lattice symmetry of monolayer $NiPS_3$, particularly the number and position of inversion centers, disallows these other contributions to the magnetic anisotropy to leading order. Hence, we do not consider them in our analysis.

Supplementary Note 3: BKT transition and the q-state clock model

The magnetic transitions of monolayer $NiPS_3$ at finite temperature can be modeled by the q-state clock model, which reads

$$H = J \sum_{\langle \mathbf{r},\mathbf{r}' \rangle} \cos[\psi(\mathbf{r}) - \psi(\mathbf{r}')] + K_{\mathsf{q}} \sum_{r} \cos(\mathsf{q}\psi(\mathbf{r})), \tag{S4}$$

where ψ is the in-plane azimuthal angle of spins in a 2D lattice. $|K_{\mathsf{q}}| < J$ is the symmetry-allowed anisotropy due to spin-lattice coupling and spin-orbit coupling, indexed by a positive integer q[8]. Following the analysis in Supplementary Note 2, the honeycomb lattice structure implies that K_6 is the dominant in-plane anisotropy. In addition, strain effects may lead to weak K_2 Ising anisotropy. Here, we consider the limit $J > |K_6| \gg |K_2|$.

Infinite-size limit: Renormalization group (RG) analyses and numerical simulations of the infinite-size system have produced broadly consistent results. RG studies have shown that for $\mathsf{q} > 4$, the system exhibits three distinct phases, separated by two phase transitions – one of which is of the BKT type[8]. In contrast, for $\mathsf{q} \leq 4$, the K_{q} term acts as a relevant perturbation at all temperatures, reducing the phase diagram to only two phases. When $\mathsf{q} = 2$, the transition falls within the Ising universality class.

Finite-size results:

Isotropic

Finite-size effects play a critical role for the system sizes relevant to experimental studies. In the isotropic limit ($K_{\mathsf{q}} = 0$ for all q), previous work[9] has shown that the magnetization M – serving as the magnetic order parameter – scales with the system area A as $M \propto (\frac{1}{2\mathsf{A}})^{1/(8\pi\kappa_{\text{eff}})}$, where κ_{eff} is the effective stiffness, which depends on both the temperature and system size. Notably, Ref. 10 demonstrated that just below the BKT transition temperature T_{BKT}, the magnetization exhibits critical behavior, scaling as $M(T) \propto (1 - T/T_{BKT})^{\beta}$, with an effective critical exponent $\beta = 3\pi^2/128 \approx 0.231$. A similar scaling behavior is expected in the antiferromagnetic case, where the magnetization is replaced by the antiferromagnetic order parameter.

K_6 anisotropy

The behavior of six-state clock models in finite-sized systems is expected to deviate from that in the thermodynamic limit. As in the infinite-size case, two distinct phase transitions are anticipated: a high-temperature BKT transition and a low-temperature transition into an ordered phase. The critical temperature separating these phases depends on the magnitude of the $\mathbb{Z}_6$ anisotropy term (K_6). Similar to the isotropic finite-sized 2D XY model, the order parameter in the BKT phase exhibits a pronounced dependence on system size. Furthermore, other physical observables are expected to vary in an intricate way with system parameters. To investigate these finite-size and anisotropy effects, we perform atomistic Monte Carlo simulations across a range of K_6 values, starting from the pure 2D XY limit ($K_6 = 0$), and plot the temperature-dependent magnetic order parameter in Fig. S19a-b.

For large values of K_6, a distinct kink appears in the temperature-dependent magnetic structure factor at intermediate temperatures, signaling the transition at T_N into the six-clock antiferromagnetic phase. As K_6 is reduced, this second transition becomes progressively less pronounced and more difficult to resolve, particularly for $K_6 < 1$ meV. Also, by focusing on the low temperature region for these low K_6 values (see Fig. S19b), we observe an approximate linear decrease in the magnetic structure factor with decreasing K_6. Notably, this trend that abruptly ends between $K_6 = 0.1$ and 0 meV, where a clear jump can be seen. This behavior is emphasized in Fig. S19c, where the data are replotted on logarithmic axes and compared to an order-parameter scaling relation using the 2D XY critical exponent β, fitted to the high-temperature regime. A pronounced deviation from the expected 2D XY scaling

emerges even at $K_6 = 0.1$ meV, indicating that signatures of both phase transitions persist down to the smallest finite K_6 explored.

This behavior is consistent with the expected system-size dependence. In the thermodynamic limit, the magnetic order parameter vanishes uniformly throughout the BKT phase, leading to a divergent contrast between the BKT and six-clock antiferromagnetic phases – irrespective of the magnitude of the K_6 anisotropy. In finite-sized systems, however, this contrast becomes finite, governed by the interplay between system size and the strength of the K_6 anisotropy.

A comparison between the simulations and experimental results – based solely on the contrast between the BKT and six-clock antiferromagnetic phases – might suggest that the experimental data are consistent with a relatively large six-fold anisotropy on the order of 1 meV. However, the simulated system size ($L = 100$) is approximately an order of magnitude smaller than the experimental probe spot size of 1 μm. While direct simulations at this experimental scale are computationally prohibitive due to convergence issues and excessive runtime, the size dependence shown in Fig. 3f reveals that, at larger system sizes, an equivalent contrast could be achieved with a K_6 value nearly one-tenth of that used in the simulations. This suggests that the true effective six-fold anisotropy in the material may lie in the hundreds of μeV range.

Finally, we note that in these finite-size simulations, the observed transition temperature T_N increases with the strength of the K_6 anisotropy (see Fig. S19d), consistent with the trend expected in the thermodynamic limit.

K_2 vs. K_6 anisotropy

The role of a weak Ising anisotropy ($K_2 \ll K_6$) is particularly important in finite-size systems. In the thermodynamic limit, K_2 acts as a relevant perturbation at all temperatures, overriding any six-state clock physics irrespective of the magnitude of the anisotropy. To investigate the finite-size case, we performed atomistic Monte Carlo simulations with varying K_2 while keeping $K_6 = 1$ meV fixed. The resulting temperature-dependent magnetic structure factors are shown in Fig. S20a-c, and the extracted critical exponents β and T_N are summarized in Fig. S20d-e. We find that increasing K_2 lowers T_N and continuously reduces the critical exponent from the 2D XY limit toward an Ising-like response, reflecting the gradual suppression of six-state clock behavior. Importantly, for sufficiently weak Ising anisotropy ($K_2 \ll K_6$), a characteristic length scale emerges near T_N below which the effects of K_2 are negligible. In this finite-size regime, the system retains the critical scaling of the BKT transition. Our experimental results lie in this limit: a small $\mathbb{Z}_2$ anisotropy biases domain selection in the low-temperature antiferromagnetic phase, yet the overall physics remain distinctly six-clock in character.

It is worth noting that in these simulations, the six-clock phase appears unusually robust to variations in K_2, persisting even when K_2 and K_6 are of comparable magnitude. This robustness is a consequence of the relatively small system size used in the simulations. As the system size approaches the thermodynamic limit, the six-clock phase is expected to become progressively more fragile. Additional large-scale simulations will therefore be required to accurately determine how the phase boundaries evolve in the experimentally relevant regime.

Supplementary Note 4: Sources of the SHG signal

In this section, we analyze the origin of the SHG response and consider the temperature-dependent SHG signal along with the order parameter fits discussed in the main text. As outlined in the main text, the total second-harmonic polarization can be decomposed into three primary contributions:

$$P_i(2\omega) \propto \chi^{ED}_{ijk} E_j(\omega) E_k(\omega) + \chi^{EQ}_{ijkl} E_j(\omega) \nabla_k E_l(\omega) + \chi^{MD}_{ijk} E_j(\omega) B_k(\omega), \tag{S5}$$

which correspond to the electric dipole (ED), electric quadrupole (EQ), and magnetic dipole (MD) contributions, respectively, to the nonlinear optical response.

Since neither the crystal structure nor the spin-lattice configuration in $NiPS_3$ breaks inversion symmetry, the ED contribution is symmetry-forbidden, aside from a weak surface-induced effect. As a result, all three mechanisms – ED (via surface effects), EQ, and MD – can contribute with comparable strength and, in principle, exhibit coupling to the magnetic order. A comprehensive decomposition of the anisotropic nonlinear response associated with each SHG mechanism is presented in Supplementary Note 5. The resulting angular patterns share formally identical Fourier components across the different mechanisms, rendering it impossible to unambiguously distinguish them based on polarimetry alone.

Probe photon energy dependence: To identify the origin of the magnetic SHG signal, we compare the SHG response from thin $NiPS_3$ both above and below T_N using two fundamental probe photon energies – 1.20 eV and 1.55 eV – as shown in Fig. S7. The SHG signal at 1.55 eV is substantially weaker and exhibits little temperature dependence across the magnetic transition, in contrast to the pronounced enhancement observed at 1.20 eV. This distinction is significant because both photon energies show comparable optical absorption (see Fig. 3b) yet they differ in resonance conditions: 1.55 eV lies on the tail of the sideband to the exciton peak, whereas 1.20 eV is resonant with a parity-forbidden d–d transition. The strong magnetic sensitivity at 1.20 eV suggests a dominant contribution from MD-type SHG processes.

Electric-dipole SHG: Neither the monolayer nor the multilayer $NiPS_3$ samples – whether strained or unstrained – intrinsically break inversion symmetry and thus do not support bulk ED SHG. However, in our experimental configuration, thin $NiPS_3$ flakes are interfaced with vacuum on one side and thermally grown SiO_2 on the other. This asymmetric dielectric environment breaks any symmetries that exchange , including any inversion, out-of-plane roto-inversion, and in-plane rotation symmetries. This reduces the point group from $\bar{3}m$ to $3m$ in the monolayer case and from $2/m$ to m in the multilayer case. Assuming a mirror plane $\sigma_m \parallel y$, this symmetry reduction permits the emergence of a static electric polarization $\mathbf{P_0}$ along z in the monolayer and within the xz plane in the multilayer system, thereby enabling ED–allowed SHG contributions.

If present, this type of SHG would exhibit two notable features: (1) it would be resonantly enhanced by ED-allowed transitions in the system, and (2) a portion of the response would scale linearly with the antiferromagnetic order, *i.e.*, $\Delta\chi^{ED} \propto \mathbf{P}_0\mathbf{L}$[11,12]. The first point implies that an ED-SHG signal should be observable when using a fundamental photon energy of 1.55 eV, which has a comparable absorption cross-section to 1.20 eV, the photon energy used in our primary measurements. However, as shown in Fig. S7, we observe no discernible temperature dependence in either the SHG intensity or its angular anisotropy at 1.55 eV. This suggests that any ED-SHG response is either weak or does not strongly couple to the magnetic order.

Moreover, if ED-SHG were the dominant mechanism, its linear dependence on the order parameter would imply critical exponents of $\beta > 0.6$ in the multilayer and $\beta > 0.4$ in the monolayer regime – values that are incompatible with any known 2D or 3D spin universality class. Taken together, these ob-

servations strongly indicate that ED-SHG does not significantly contribute to the temperature-dependent magnetic SHG signal we observe.

Electric-quadrupole SHG: The EQ contribution is a nonresonant SHG process that does not require inversion symmetry breaking and is typically associated with a bulk response, often observed in much thicker samples than those investigated here. While EQ SHG can, in certain cases, couple to the magnetic order parameter, its nonresonant nature implies that it should be present for both 1.55 eV and 1.20 eV excitation energies – much like the ED contribution. However, as with ED SHG, we observe no temperature-dependent changes in the SHG signal at 1.55 eV (Fig. S7), suggesting that EQ SHG does not significantly contribute to the magnetic SHG signal in our measurements.

Magnetic-dipole SHG: The considerations above leave us with MD SHG as the most plausible origin of the observed signal. By tuning the fundamental probe photon energy to a spin-allowed but parity-forbidden d–d transition – specifically near $\sim$1.20 eV – we gain access to MD SHG, which is highly sensitive to magnetostrictive effects in the material. This mechanism has been shown to reliably track magnetic order in other Ni-based antiferromagnets with centrosymmetric crystal and spin structures[13]. In our case, the targeted d–d transition is the same as that used to study NiO[13], but is slightly blue-shifted due to the different ligand environment (S^{2-} in $NiPS_3$ vs. O^{2-} in NiO) and the presence of a small trigonal distortion. Moreover, it has been demonstrated that the angular anisotropy of the MD SHG response closely follows the underlying magnetic domain structure in such materials, further supporting its role as the dominant SHG mechanism in our system.

Temperature-dependent SHG signal: Based on these observations, we conclude that the ED and EQ contributions are both weak and largely insensitive to temperature and the onset of antiferromagnetic order, consistent with previous reports[14]. In contrast, the pronounced SHG signal observed at 1.20 eV can be primarily attributed to MD SHG arising from magnetostriction, which is expected to scale with the square of the magnetic order parameter l, *i.e.*, $\chi^{MD}_{ijk} \propto l^2$ (see Supplementary Notes 6 and 7).
Near the critical temperature T_c, the order parameter scales as $l \propto (1 - T/T_c)^{\beta}$, where β is the critical exponent. Substituting this into the nonlinear susceptibility expansion, the SHG intensity becomes

$$I_{SHG}(T) = \left|\tilde{\chi}^{ED} + \tilde{\chi}^{EQ} + \tilde{\chi}^{MD,(0)} + \tilde{\chi}^{MD,(2)} l^2 + O\left(l^4\right)\right|^2 \tag{S6}$$

$$= |\tilde{\chi}_0|^2 + 2\,\mathrm{Re}\left[\tilde{\chi}_0^* \tilde{\chi}^{MD,(2)}\right](1 - T/T_c)^{2\beta} + O\left((1 - T/T_c)^{4\beta}\right), \tag{S7}$$

where $\tilde{\chi}_0 = \tilde{\chi}^{ED} + \tilde{\chi}^{EQ} + \tilde{\chi}^{MD,(0)}$ is the temperature-independent background second-order nonlinear susceptibility from residual ED, EQ, and non-critical MD contributions.

To reliably extract the critical exponents, we fit the square root of the SHG intensity across the entire temperature region with a nonlinear least-squares routine using the following order-parameter model:

$$I^{1/2}_{SHG}(T) = |\tilde{\chi}_0| + \chi^{MD}_2 (1 - T/T_c)^{2\beta}, \tag{S8}$$

where $\chi^{MD}_2 = \mathrm{Re}(\tilde{\chi}_0^* \tilde{\chi}^{MD,(2)})/|\tilde{\chi}_0|$. Note that this truncation implies $|\tilde{\chi}^{MD,(2)}|^2 (1 - T/T_c)^{2\beta} \ll \mathrm{Re}(\tilde{\chi}_0^* \tilde{\chi}^{MD,(2)})$. Fitting Eq.(S8) to the temperature-dependent SHG data shown in Fig.S17, which underlie the data in Fig.3c,d, yields the parameters listed in TableS1. We also perform an alternative fit in which $|\tilde{\chi}_0|$ is fixed to the average high-temperature SHG amplitude as reported in Table S3; both fitting procedures return consistent values for T_c and β. In all cases, because the model is fitted globally over all relevant data – both above and below T_c – the resulting error bars represent a statistically rigorous estimate of the underlying experimental parameter uncertainty.

As an additional assessment of the parameter uncertainties, we perform a bootstrap analysis using the same nonlinear fitting model. In each bootstrap cycle, a new dataset of equal size is generated by resampling with replacement from the original measurements, and Eq. (S8) is refitted to obtain a distribution of parameters. The mean values and 95% confidence intervals extracted from this distribution are summarized in Table S2. The bootstrap results agree well with the fits discussed above, although the confidence intervals are somewhat broader. This is expected: bootstrap resampling assumes a simplified error model and can yield unrepresentative fits when points near the transition temperature are randomly omitted, producing long parameter tails that inflate the 95% confidence bounds. Despite this, the overall consistency of all fitting approaches demonstrates the robustness of the extracted critical parameters.

Supplementary Note 5: SHG polarimetry

In this section, we analyze the anisotropic SHG responses allowed by the crystal symmetries. We adopt the same coordinate convention used previously, with the x and y axes defined as in-plane directions and z as the out-of-plane axis. Additionally, we assume that the two-fold rotation axis (C_2') and the normal to the mirror plane (m) are both aligned along the y-axis. ψ is the in-plane azimuthal angle measured from the x axis. The in-plane polarization axis ϕ is measured from the y-axis and is related to ψ by $\phi = \psi + \pi/2$.

Magnetic-dipole SHG: The symmetry-allowed tensor elements for MD SHG in the monoclinic $2/m$ point group, relevant for multilayer $NiPS_3$, are

$$\tilde{\chi}^{MD}_{yyy}, \tilde{\chi}^{MD}_{xxy}, \tilde{\chi}^{MD}_{xyx}, \tilde{\chi}^{MD}_{yxx},$$

which give rise to the following components of the nonlinear polarization:

$$\begin{aligned} P^{MD}_{XY} &= \tilde{\chi}^{MD}_{xyx} \sin^3\psi + (\tilde{\chi}^{MD}_{yyy} - \tilde{\chi}^{MD}_{yxx} - \tilde{\chi}^{MD}_{xxy}) \cos^2\psi \sin\psi, \\ P^{MD}_{XX} &= \tilde{\chi}^{MD}_{xxy} \cos^3\psi + (\tilde{\chi}^{MD}_{yyy} - \tilde{\chi}^{MD}_{xyx} - \tilde{\chi}^{MD}_{yxx}) \cos\psi \sin^2\psi. \end{aligned} \tag{S9}$$

Imposing the monolayer $\bar{3}m$ symmetry, we obtain: $\tilde{\chi}^{MD}_{xxy} = \tilde{\chi}^{MD}_{xyx} = \tilde{\chi}^{MD}_{yxx} = -\tilde{\chi}^{MD}_{yyy}$, which leads to the following expressions for the nonlinear polarization components:

$$\begin{aligned} P^{MD}_{XY} &= -\tilde{\chi}^{MD}_{xxy} \sin(3\psi), \\ P^{MD}_{XX} &= \tilde{\chi}^{MD}_{xxy} \cos(3\psi). \end{aligned} \tag{S10}$$

This implies that, at high temperatures, the SHG response of the monolayer should exhibit a fully isotropic or six-fold symmetric pattern, with a two-fold anisotropy emerging only upon the onset of rotational symmetry breaking at low temperatures. This expectation is broadly consistent with the behavior observed in the cross-polarized configuration, which remains largely isotropic at high temperature. However, a weak two-fold anisotropy persists in the parallel-polarized configuration even above the transition, suggesting a subtle symmetry breaking already present in that geometry. We attribute this residual anisotropy to strain introduced during the exfoliation process, which distorts the high-temperature polar pattern and biases the system toward a preferred spin-chain direction in the six-clock phase. Such strain effects are theoretically compatible with the six-clock model, as supported by Monte Carlo simulations and symmetry considerations presented in Supplementary Notes 3 and 4. Finally, we note that the temperature dependence of the anisotropy in the parallel-polarized configuration mimics that observed in the cross-polarized channel, consistent with the expectations of the magnetostrictive mechanism described in Supplementary Note 6.

Electric-dipole SHG: Due to the inversion symmetry inherent in both the $\bar{3}m$ and $2/m$ point groups, bulk ED SHG is forbidden, and only a surface-induced ED contribution remains. For the reduced surface symmetry described by the point groups $3m$ and m, the allowed ED susceptibility tensor elements at normal incidence are:

$$\tilde{\chi}^{ED}_{xxx}, \tilde{\chi}^{ED}_{xyy}, \tilde{\chi}^{ED}_{yyx} = \tilde{\chi}^{ED}_{yxy},$$

which yield the following expressions for the nonlinear polarization components:

$$\begin{aligned} P^{ED}_{XY} &= -\tilde{\chi}^{ED}_{xyy} \sin^3\psi + (2\tilde{\chi}^{ED}_{yyx} - \tilde{\chi}^{ED}_{xxx}) \cos^2\psi \sin\psi, \\ P^{ED}_{XX} &= \tilde{\chi}^{ED}_{xxx} \cos^3\psi + (2\tilde{\chi}^{ED}_{yyx} + \tilde{\chi}^{ED}_{xyy}) \cos\psi \sin^2\psi. \end{aligned} \tag{S11}$$

Electric-quadrupole SHG: The symmetry-allowed tensor elements for EQ SHG in the $2/m$ point group are

$$\chi^{EQ}_{xxzx}, \chi^{EQ}_{yyzx}, \chi^{EQ}_{yxzy}, \chi^{EQ}_{xyzy},$$

which give rise to the following expressions for the nonlinear polarization components:

$$\begin{aligned} P^{EQ}_{XY} &= -\chi^{EQ}_{xyzy}\sin^3\psi + (\chi^{EQ}_{yyzx} + \chi^{EQ}_{yxzy} - \chi^{EQ}_{xxzx})\cos^2\psi\sin\psi, \\ P^{EQ}_{XX} &= \chi^{EQ}_{xxzx}\cos^3\psi + (\chi^{EQ}_{yyzx} + \chi^{EQ}_{yxzy} + \chi^{EQ}_{xyzy})\cos\psi\sin^2\psi. \end{aligned} \tag{S12}$$

Total SHG: The overall SHG signal, expressed as a function of the in-plane polarization angle $\phi = \psi + \pi/2$ and used throughout the main text and figures, is given by:

$$\begin{aligned} I_{XY}(\phi) &= A_2\cos^2(\phi) + A_4\cos^4(\phi) + A_6\cos^6(\phi), \\ I_{XX}(\phi) &= B_2\sin^2(\phi) + B_4\sin^4(\phi) + B_6\sin^6(\phi), \end{aligned} \tag{S13}$$

where

$$\begin{aligned} A_6 &= |\tilde{\chi}_t|^2, \\ A_4 &= -2|\tilde{\chi}_t|[|\tilde{\chi}_t|\cos\theta_A - |\tilde{\chi}_A|], \\ A_2 &= |\tilde{\chi}_A|^2 + |\tilde{\chi}_t|^2 - 2|\tilde{\chi}_t||\tilde{\chi}_A|\cos\theta_A, \\ B_6 &= |\tilde{\chi}_t|^2, \\ B_4 &= -2|\tilde{\chi}_t|[|\tilde{\chi}_t|\cos\theta_B - |\tilde{\chi}_B|], \\ B_2 &= |\tilde{\chi}_B|^2 + |\tilde{\chi}_t|^2 - 2|\tilde{\chi}_t||\tilde{\chi}_B|\cos\theta_B, \end{aligned}$$

and

$$\begin{aligned} \tilde{\chi}_t &= \tilde{\chi}^{MD}_{xxy} + \tilde{\chi}^{MD}_{xyx} + \tilde{\chi}^{MD}_{yxx} - \tilde{\chi}^{MD}_{yyy} + \tilde{\chi}^{ED}_{xxx} - 2\tilde{\chi}^{ED}_{yyx} - \tilde{\chi}^{ED}_{xyy} + \tilde{\chi}^{EQ}_{xxzx} - \tilde{\chi}^{EQ}_{yyzx} - \tilde{\chi}^{EQ}_{yxzy} - \tilde{\chi}^{EQ}_{xyzy}, \\ \tilde{\chi}_A &= \tilde{\chi}^{MD}_{xyx} - \tilde{\chi}^{ED}_{xyy} - \tilde{\chi}^{EQ}_{xyzy}, \\ \tilde{\chi}_B &= \tilde{\chi}^{MD}_{xxy} + \tilde{\chi}^{ED}_{xxx} + \tilde{\chi}^{EQ}_{xxzx}, \\ \theta_A &= \arg[\tilde{\chi}_t(\tilde{\chi}_A)^*], \\ \theta_B &= \arg[\tilde{\chi}_t(\tilde{\chi}_B)^*]. \end{aligned}$$

Since all three SHG mechanisms – ED, EQ, and MD – contribute to the same Fourier components in the anisotropic SHG signal, polarimetry alone cannot uniquely identify the origin of the observed response. Additionally, due to dissipative effects in the system, the complex phase of the susceptibility terms cannot be reliably used to distinguish between them. Disentangling these contributions therefore requires additional experimental input, such as the probe photon energy dependence discussed in Supplementary Note 4 and Fig. S7.

Finally, we note that by neglecting both ED and EQ contributions and expanding the MD susceptibility to second order in the magnetic order parameter l, one recovers the simplified expression presented in Supplementary Note 2.

Resonant SHG polarimetry: Here, we analyze the potential influence of the resonant SHG process on the observed polarimetry patterns. For a given resonant SHG mechanism, the nonlinear susceptibility can be expressed in terms of a product of transition dipole moments connecting the involved electronic

states ψ_1, ψ_2, ψ_3:

$$\tilde{\chi}_{ijk} \propto \frac{\langle\psi_1|\,\boldsymbol{\mu}_i\,|\psi_3\rangle\,\langle\psi_3|\,\boldsymbol{\mu}_j\,|\psi_2\rangle\,\langle\psi_2|\,\boldsymbol{\mu}_k\,|\psi_1\rangle}{(2\omega-\omega_{31})(\omega-\omega_{21})}, \tag{S14}$$

where $\boldsymbol{\mu}_i$ denotes the electric or magnetic dipole operator along the i-th coordinate. Any anisotropic contribution to the SHG signal thus depends on the symmetry properties of the dipole operators under the point group of the system.

We first consider the MD SHG process, which involves three steps: a MD transition from the ground state $^3\mathrm{A}_{2\mathrm{g}}$ to an intermediate $^3\mathrm{E}_{\mathrm{g}}$ state, followed by an ED transition to a higher-lying electronic band of u symmetry, and finally, an ED transition back to the $^3\mathrm{A}_{2\mathrm{g}}$ ground state. The MD transition corresponds to an on-site d–d excitation, and we analyze it within the $\bar{3}m$ point group symmetry of the NiS_6 octahedra, slightly perturbed by trigonal distortion. In this symmetry, the MD operators (M_x, M_y) transform according to $^3\mathrm{E}_{\mathrm{g}}$, which does not permit anisotropy in our experimental geometry. Similarly, the in-plane ED operators (E_x, E_y) transform as components of E_{u} under $\bar{3}m$, and thus also do not introduce anisotropy in the monolayer response.

In contrast, for bulk and multilayer samples, the symmetry is that of monoclinic $2/m$, where all magnetic and electric dipole operators transform independently. In this symmetry, anisotropic SHG contributions become allowed, and the ED transitions can modify the angular dependence of the SHG signal – even in resonant conditions. This may account for some of the differences observed in the polarimetry patterns between monolayer and multilayer samples. However, it is worth noting that the monoclinic distortion is relatively weak, as indicated by the absence of resolvable monoclinic splitting in optical absorption spectra[15].

Crucially, these symmetry-based anisotropic effects do not affect the monolayer signal, where we observe clear signatures of BKT physics.

Supplementary Note 6: Magnetostrictive origin of MD SHG

Symmetry analysis: In this section, we focus on a description of the SHG signal by considering the most general response functionals allowed by the symmetries of the system near the magnetic-ordering phase transition (where the order parameter is small). Let us focus on SHG of MD origin, which would follow from a response free-energy functional of the form:

$$F = -\sum_{ijk} \chi^{MD}_{ijk} E_i(-2\omega) E_j(\omega) B_k(\omega), \tag{S15}$$

allowing an induced polarization at 2ω in response to an external electromagnetic field (E_j, B_k) of frequency ω, $P_i(2\omega) = \sum_{jk} \chi^{MD}_{ijk} E_j(\omega) B_k(\omega)$. Here χ^{MD}_{ijk} is the MD SHG response tensor. As the point group symmetries of bulk and monolayer $NiPS_3$ are distinct, we begin by analyzing χ^{MD}_{ijk} for the bulk case.

The crystalline point group symmetries of bulk $NiPS_3$ are (i) a 180° rotation C_2' about the y-axis and (ii) a reflection σ_v about the xz mirror plane (see Fig. S5). Under the action of these symmetries, $E_{x/z}, B_{x/z} \xrightarrow{C_2'} -E_{x/z}, -B_{x/z}$ and $E_y, B_{x/z} \xrightarrow{\sigma_v} -E_y, -B_{x/z}$, whereas the other components remain invariant. The spins are known to be Ising-like due to an easy-axis anisotropy along the x direction. The order parameter l_x of the zigzag antiferromagnetic state, defined on 4-sublattice magnetic unit cell (illustrated with a rectangle in Fig. S5), is $l_x = \frac{1}{4}(m_x^A + m_x^B - m_x^C - m_x^D)$, where m_x^α is the magnetization at the sublattice site α. Since $m_x^\alpha \xrightarrow{C_2'} -m_x^\alpha$ and $m_x^{A/B} \xrightarrow{\sigma_v} -m_x^{C/D}$, the transformations of l_x under the lattice symmetries are

$$\begin{aligned} l_x &\xrightarrow{C_2'} -l_x, \\ l_x &\xrightarrow{\sigma_v} l_x. \end{aligned} \tag{S16}$$

We now organize χ^{MD}_{ijk} by expressing it as a polynomial in l_x: $\chi^{MD}_{ijk} = \chi^{MD,(0)}_{ijk} + \chi^{MD,(1)}_{ijk} l_x + \chi^{MD,(2)}_{ijk} l_x^2$, and dropping higher order terms which are small near the phase transition. By demanding that F remains invariant under the crystal symmetries, we find that for even polynomial order n, the following coefficients are non-zero and independent of each other:

$$\chi^{MD,(n)}_{xxy}, \chi^{MD,(n)}_{yyy}, \chi^{MD,(n)}_{zzy}, \chi^{MD,(n)}_{xzy}, \chi^{MD,(n)}_{xyx}, \chi^{MD,(n)}_{xyz}, \chi^{MD,(n)}_{yzx}, \chi^{MD,(n)}_{yzz} \neq 0, \tag{S17}$$

as well as the coefficients corresponding to exchanging the first two labels. However, crucially, all the coefficients with odd n are zero. Therefore, the leading order dependence of χ^{MD} on l_x is quadratic near the phase transition:

$$I_{SHG} \propto \text{constant} + A \cdot l_x^2. \tag{S18}$$

We note that the allowed MD tensor elements are identical to the third-order axial i- and c-tensors for the point group $2/m$, which are, in turn, identical to the ED SHG for the reduced point group 2_y [16].

Let us now consider the monolayer case. In addition to the C_2' and σ_v symmetries, monolayer $NiPS_3$ has a C_{3z} rotational symmetry about the z-axis. The spin symmetry also differs from the bulk case due to the absence of easy-axis anisotropy along x. Instead, a six-fold anisotropic term $-\cos(6\psi_i)$, consistent with the crystal symmetries, can emerge in the Hamiltonian, where ψ_i denotes the angle of the spin moment at site i relative to the x-axis. This term energetically favors locking spins into one of six degenerate minima in the magnetically ordered phase, dramatically altering the critical properties of the magnetic phase transition [8].

We define the antiferromagnetic order parameter in terms of the four magnetic moments in the magnetic unit cell $\vec{l} = \frac{1}{4}(\vec{m}_A + \vec{m}_B - \vec{m}_C - \vec{m}_D)$, ensuring that $\vec{l}$ transforms as a vector under C_{3z}, unlike the Ising spins in the bulk case. We also define $l \equiv |\vec{l}|$. To determine whether there are linear-in-l contributions to the SHG tensor – and consequently to the free energy – we first examine the possible terms of the form:

$$F = \Gamma^{(1)}_{ijkl} E_i(-2\omega) E_j(\omega) B_k(\omega) l_l, \tag{S19}$$

where $\Gamma^{(n)}_{ijk...}$ is a generalized susceptibility tensor corresponding to a free-energy functional of order $O(l^n)$. The indices are to be contracted to yield products of the following C_{3z} invariant terms: $\vec{l} \cdot \vec{E}$, $\hat{z} \cdot \vec{l} \times \vec{E}$, $\vec{l} \cdot \vec{B}$, $\hat{z} \cdot \vec{l} \times \vec{B}$, E_z^2 and $E_z B_z$. We find that none of such contractions satisfy the C_2' and σ_v symmetries of the crystal, and hence contributions linear-in-l are disallowed. Now we turn to quadratic-in-l contributions arising from a free energy of the form $\Gamma^{(2)}_{ijklm} E_i(-2\omega) E_j(\omega) B_k(\omega) l_l l_m$. Analogous to the linear case, we decompose the terms in the free energy into products of C_{3z} invariant pieces to obtain the following symmetry-allowed terms:

$$\begin{aligned} F = &- \hat{z} \cdot \vec{l} \times \vec{E} (\chi_1^{MD} (\vec{l} \cdot \vec{E}) B_z + \chi_2^{MD} (\vec{l} \cdot \vec{B}) E_z) \\ &- E_z (\chi_3^{MD} (\vec{l} \cdot \vec{E}) (\hat{z} \cdot \vec{l} \times \vec{B}) + \chi_4^{MD} l^2 (\hat{z} \cdot \vec{E} \times \vec{B})). \end{aligned} \tag{S20}$$

This suggests that near the phase transition on the ordered side, I_{SHG} remains quadratic in l.

Electro- and magneto-striction: We now turn from the symmetry-allowed contributions to MD SHG to a discussion of their possible microscopic origins. For the induced polarization at 2ω to be sensitive to the magnetic order parameter, the two quantities must be coupled. Microscopic mechanisms that enable this coupling are electro- and magneto-striction, where the elastic strain in the crystal ϵ_{ij} interacts with the electric polarization $\vec{P}$ and the magnetic order parameter $\vec{l}$, respectively. As a result, strain can mediate an indirect coupling between $\vec{P}$ and $\vec{l}$. In this section, we explicitly analyze the nature of these couplings and how they lead to the terms discussed earlier.

We begin by examining the elastic, electro- and magneto-strictive contributions to the Lagrangian density that respect the crystal symmetries:

$$\begin{aligned} \mathcal{L}_{el} &= \lambda_1^{el} \epsilon_{ij}^2 + \lambda_2^{el} \epsilon_{ii}^2 + \lambda_3^{el} \epsilon_{ix} \epsilon_{iz} + \lambda_4^{el} \epsilon_{ii} \epsilon_{xz}, \\ \mathcal{L}_{es} &= \lambda_1^{es} \epsilon_{ij} P_i P_j + \lambda_2^{es} \epsilon_{ii} P^2 + \lambda_3^{es} \epsilon_{ix} P_i P_z + \lambda_4^{es} \epsilon_{iz} P_i P_x + \lambda_5^{es} \epsilon_{ii} P_x P_z + \lambda_6^{es} \epsilon_{xz} P^2, \\ \mathcal{L}_{ms} &= \lambda_1^{ms} \epsilon_{ij} l_i l_j + \lambda_2^{ms} \epsilon_{ii} l^2 + \lambda_3^{ms} \epsilon_{ix} l_i l_z + \lambda_4^{ms} \epsilon_{iz} l_i l_x + \lambda_5^{ms} \epsilon_{ii} l_x l_z + \lambda_6^{ms} \epsilon_{xz} l^2. \end{aligned} \tag{S21}$$

Here, we define $P^2 = P_i^2$ with implicit summation over repeated indices. The couplings $\lambda_{i>2}$ are not C_{3z} symmetric and therefore vanish in the monolayer case. Optimizing with respect to ϵ by setting $\frac{\delta \mathcal{L}}{\delta \epsilon_{ij}} = 0$ constrains ϵ to be quadratic in l (assuming $P = 0$ in equilibrium). Substituting this result into $\mathcal{L}_{es}$ then yields the following coupling:

$$\mathcal{L}_{l-P} = \kappa_1 (\vec{l} \cdot \vec{P})^2 + \kappa_2 l^2 P^2 + \vec{l} \cdot \vec{P} \left(\kappa_3 l_x P_z + \kappa_4 l_z P_x\right) + \kappa_5 l^2 P_x P_z + \kappa_6 l_x l_z P^2. \tag{S22}$$

As before, $\kappa_{i>2} = 0$ for the monolayer. We now introduce the remaining terms in the total Lagrangian

density, $\mathcal{L}_{tot}$, accounting for the presence of an external electromagnetic field with frequency ω.

$$\begin{aligned}
\mathcal{L}_{tot} &= \mathcal{L}_l + \mathcal{L}_P + \mathcal{L}_{l-P} + \mathcal{L}_{l-B} + \mathcal{L}_{P-E}, \\
\mathcal{L}_l &= \frac{1}{2}\left(\partial_t \vec{l}\right)^2 + r^l l^2 + u^l l^4 + \mathcal{L}_l^{ani.}, \\
\mathcal{L}_P &= \frac{1}{2}\left(\partial_t \vec{P}\right)^2 + r_P P^2, \\
\mathcal{L}_{l-B} &= \alpha_1^l l^2 B_y + \alpha_2^l l_x l_y B_x + \alpha_3^l l_x l_y B_z + \alpha_4^l l_z \hat{z} \cdot \vec{l} \times \vec{B}, \\
\mathcal{L}_{P-E} &= \alpha^P \vec{P} \cdot \vec{E}.
\end{aligned} \tag{S23}$$

Here, we have omitted the spatial gradient terms, as we are primarily concerned with zero-momentum responses. $\mathcal{L}_l^{ani.}$ represents the spin-anisotropy term, which differs between the bulk and the monolayer:

$$\begin{aligned}
\text{Bulk: } \mathcal{L}_l^{ani.} &= -k_2^l l_x^2. \\
\text{Monolayer: } \mathcal{L}_l^{ani.} &= -k_6^l \cos(6\psi).
\end{aligned} \tag{S24}$$

Here, ψ is the azimuthal angle of $\vec{l}$ relative to the x-axis, and l_x is the x-component of $\vec{l}$. We choose $r_l < 0$ and $r_P > 0$ to select the ground state where $\vec{l} \neq 0$ and $\vec{P} = 0$. $\mathcal{L}_{l-B}$ and $\mathcal{L}_{P-E}$ describe the leading couplings of the order parameters to the external electromagnetic field. The coupling between $\vec{l}$ and the external magnetic field occurs at quadratic order in $\vec{l}$, in contrast to the direct linear coupling in the case of ferromagnets. Note that the first three coupling types, with strength α_1^l-α_3^l, are not C_{3z} symmetric and are therefore forbidden for the monolayer, being non-zero only for the bulk.

Having formulated the total Lagrangian density, the next step is to derive the equations of motion in order to calculate the response of $\vec{l}$ and $\vec{P}$ to the applied external electromagnetic field. The equation motion for $\vec{P}$ is

$$\begin{aligned}
\partial_t^2 \vec{P} = {} & 2r^P \vec{P} + \vec{l}(2\kappa_1 \vec{l} \cdot \vec{P} + \kappa_3 l_x P_z + \kappa_4 l_z P_x) + l^2(2\kappa_2 \vec{P} + \kappa_5 P_x \hat{z} + \kappa_5 P_z \hat{x}) + \\
& + \vec{l} \cdot \vec{P}(\kappa_3 l_x \hat{z} + \kappa_4 l_z \hat{x}) + 2\kappa_6 l_x l_z \vec{P} + \alpha^P \vec{E}.
\end{aligned} \tag{S25}$$

Expressing $\vec{P}(t) = \vec{P}^1 e^{-i\omega t} + \vec{P}^2 e^{-i2\omega t}$, $\vec{l}(t) = \vec{l}^0 + \vec{l}^1 e^{-i\omega t} + \vec{l}^2 e^{-i2\omega t}$, $\vec{E} = \vec{E}^1 e^{-i\omega t}$ and solving the equation of motion order by order in $e^{-i\omega t}$, we obtain

$$\vec{P}^1 \sim \alpha^P \vec{E}^1, \tag{S26}$$

$$\begin{aligned}
\vec{P}^2 \sim {} & \vec{l}^0 \left(2\kappa_1 \vec{l}^1 \cdot \vec{P}^1 + \kappa_3 l_x^1 P_z^1 + \kappa_4 l_z^1 P_x^1\right) + \vec{l}^1 \left(2\kappa_1 \vec{l}^0 \cdot \vec{P}^1 + \kappa_3 l_x^0 P_z^1 + \kappa_4 l_z^0 P_x^1\right) + \\
& + \vec{l}^0 \cdot \vec{l}^1 \left(2\kappa_2 \vec{P}^1 + \kappa_5 P_x^1 \hat{z} + \kappa_5 P_z^1 \hat{x}\right) + \vec{l}^0 \cdot \vec{P}^1 \left(\kappa_3 l_x^1 \hat{z} + \kappa_4 l_z^1 \hat{x}\right) \\
& + \vec{l}^1 \cdot \vec{P}^1 \left(\kappa_3 l_x^0 \hat{z} + \kappa_4 l_z^0 \hat{x}\right) + 2\kappa_6 \vec{P}^1 \left(l_x^0 l_z^1 + l_x^1 l_z^0\right).
\end{aligned} \tag{S27}$$

The dependence of $\vec{l}^1$ on the external fields will differ between the bulk and the monolayer due to the distinct forms of $\mathcal{L}_{l-B}$ in each case. Let us first focus on the bulk scenario. In this case, the equation of motion for $\vec{l}$ is

$$\begin{aligned}
\partial_t^2 \vec{l} = {} & 2r^l \vec{l} - 2k_2^l l_x \hat{x} + 4u^l l^2 \vec{l} + 2\alpha_1^l \vec{l} B_y + \left(\alpha_2^l B_x + \alpha_3^l B_z\right)(l_y \hat{x} + l_x \hat{y}) \\
& + \alpha_4^l \left(\hat{z} \cdot \vec{l} \times \vec{B} \hat{z} + l_z \vec{B} \times \hat{z}\right).
\end{aligned} \tag{S28}$$

Solving in the same manner for $\vec{P}$, with $\vec{l}^0 = l^0 \hat{x}$, we obtain

$$\vec{l}^1 \sim 2\alpha_1^l l^0 B_y \hat{x} + (\alpha_2^l B_x + \alpha_3^l B_z) l^0 \hat{y} + \alpha_4^l l^0 B_y \hat{z}. \tag{S29}$$

Substituting this into Eq. (S28) gives

$$\begin{aligned}
\chi^{MD,(2)}_{xxy} &\sim 4\alpha^l_1(\kappa_1+\kappa_2)+2\alpha^l_4(\kappa_4+\kappa_6),\\
\chi^{MD,(2)}_{yyy} &\sim 4\kappa_2\alpha^l_1,\\
\chi^{MD,(2)}_{zzy} &\sim 4\kappa_2\alpha^l_1+2\kappa_3(\alpha^l_4+\alpha^l_1),\\
\chi^{MD,(2)}_{xzy} &\sim \kappa_1\alpha^l_4+2\alpha^l_1(2\kappa_3+\kappa_5),\\
\chi^{MD,(2)}_{xyx} &\sim \kappa_1\alpha^l_2,\\
\chi^{MD,(2)}_{xyz} &\sim \kappa_1\alpha^l_3,\\
\chi^{MD,(2)}_{yzx} &\sim \kappa_3\alpha^l_2,\\
\chi^{MD,(2)}_{yzz} &\sim \kappa_3\alpha^l_3.
\end{aligned} \tag{S30}$$

Importantly, the linear-in-l terms vanish, i.e., $\chi^{MD,(1)}_{ijk}=0$, which ensures that the leading dependence of χ^{MD} and I_{SHG} on the antiferromagnetic order parameter, l, is quadratic, as demonstrated in the previous section. Furthermore, the electro- and magnetostriction mechanisms discussed here generate all the terms that are allowed by symmetry in Eq. (S17).

Repeating the same analysis for the monolayer, assuming $\vec{l}^0$ lies in the xy plane along one of the six energetically favored directions, we find

$$\vec{l}^1 \sim \alpha^l_4 \hat{z}\cdot\vec{l}\times\vec{B}\hat{z}, \tag{S31}$$

which gives

$$\vec{P}^2 \sim 2\kappa_1\alpha^l_4\left(\hat{z}\cdot\vec{l}\times\vec{B}\right)\left(\vec{l}^0\cdot\vec{E}^1\hat{z}+\vec{l}^0E^1_z\right). \tag{S32}$$

Therefore, $\chi^{MD}_3 \sim 2\kappa_1\alpha^l_4$ in Eq. (S20). Notably, for the monolayer, only χ^{MD}_3 is generated from the electro- and magnetostrictive terms considered here. Exploring other microscopic mechanisms that could generate all the symmetry-allowed coefficients will be the subject of future work.

Supplementary Note 7: Symmetry analysis of the magnetic SHG anisotropy

In this section, we examine the SHG anisotropy and its dependence on the magnetic order parameter, both above and below T_c, for monolayer and multilayer $NiPS_3$.

General considerations: The MD contribution of the SHG reads

$$P_i(2\omega) = \sum_{j,k} \tilde{\chi}^{MD}_{ijk} E_j(\omega) B_k(\omega). \tag{S33}$$

Under a unitary transformation R, let the transformation of the electric field $\vec{E}$ be denoted as $\mathcal{O}^E_{\mathsf{R}}$, and the transformation of the magnetic field $\vec{B}$ as $\mathcal{O}^B_{\mathsf{R}}$. We then have $\mathcal{O}^E_{\mathsf{R}\,ii'} P_{i'} = \sum \tilde{\chi}_{ijk} \mathcal{O}^E_{\mathsf{R}\,jj'} E_{j'} \mathcal{O}^B_{\mathsf{R}\,kk'} B_{k'}$. If R is a symmetry of the system, the SHG tensor χ_{ijk} satisfies

$$\tilde{\chi}^{MD}_{i'j'k'} = \sum \left(\mathcal{O}^E_{\mathsf{R}}\right)^{\mathsf{T}}_{i'i} \tilde{\chi}^{MD}_{ijk} \mathcal{O}^E_{\mathsf{R}\,jj'} \mathcal{O}^B_{\mathsf{R}\,kk'} = \sum \tilde{\chi}^{MD}_{ijk} \mathcal{O}^E_{\mathsf{R}\,ii'} \mathcal{O}^E_{\mathsf{R}\,jj'} \mathcal{O}^B_{\mathsf{R}\,kk'}, \tag{S34}$$

where ^{T} is the transpose operation. The unitary symmetries of interest include a two-fold rotation along the nearest Ni-Ni bond C_2', a vertical mirror normal to the nearest Ni-Ni bond σ_v, a three-fold rotation about the z-axis C_{3z}. Under these transformations, $\mathcal{O}^E$ and $\mathcal{O}^B$ read

$$\mathcal{O}^E_{C_{3z}} = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \mathcal{O}^E_{\sigma_v} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \mathcal{O}^E_{C_2'} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \tag{S35}$$

where ψ is an in-plane azimuthal rotation angle, and

$$\mathcal{O}^B_{C_{3z}} = \mathcal{O}^E_{C_{3z}}, \mathcal{O}^B_{\sigma_v} = -\mathcal{O}^E_{\sigma_v}, \mathcal{O}^B_{C_2'} = \mathcal{O}^E_{C_2'}, \tag{S36}$$

The symmetry-allowed SHG tensor components $\tilde{\chi}_{ijk}$ above T_c can be derived through Eqs. (S35) and (S36). Below T_c, $\tilde{\chi}_{ijk}$ also depends on the magnetization, necessitating an analysis of the symmetry transformation of the magnetization. We provide the details of this analysis below.

The amplitude of the experimentally measured SHG signal is proportional to $\hat{e}^E_{P/S,o} \cdot \langle \vec{P}^{2\omega} \rangle$, where $\hat{e}^E_{P/S,o}$ is the polarization of the outgoing electric field. The polarization vectors of the incoming and outgoing electric field are

$$\begin{aligned} \hat{e}^E_{P,\mathrm{i}} &= (\sin\theta\cos\psi, \sin\theta\sin\psi, -\cos\theta)\,, \ \hat{e}^E_{S,\mathrm{i}} = (\sin\psi, -\cos\psi, 0)\,, \\ \hat{e}^E_{P,o} &= (-\sin\theta\cos\psi, -\sin\theta\sin\psi, -\cos\theta)\,, \ \hat{e}^E_{S,o} = (\sin\psi, -\cos\psi, 0)\,, \end{aligned} \tag{S37}$$

where θ and ψ denote the polar and azimuthal angles, respectively. Notably, the magnetic field is oriented perpendicular to the electric field, i.e. $\hat{e}^B_{P/S,\mathrm{i}} = \hat{e}^E_{S/P,\mathrm{i}}$.

The SHG amplitude is proportional to

$$\mathcal{A}_{\mathrm{SHG}}(\theta, \psi, \mu\nu) \sim \sum_{\alpha\gamma\delta,\pm} \tilde{\chi}^{MD}_{ijk} \hat{e}^E_{\mu o,i} \hat{e}^B_{\nu \mathrm{i},j} \hat{e}^E_{\nu \mathrm{i},k} \tag{S38}$$

$$= \sum_{\alpha\gamma\delta,\pm} \tilde{\chi}^{MD}_{ijk} \hat{e}^E_{\mu o,i} \hat{e}^E_{\bar{\nu} \mathrm{i},j} \hat{e}^E_{\nu \mathrm{i},k}, \tag{S39}$$

where $\mu, \nu = P, S$, and $\bar{\nu} \neq \nu$. Below, we apply the above analysis to multilayer and monolayer $NiPS_3$.

Multilayer: Above T_c, the system retains the symmetries C_2', σ_v, and the lattice translations $\mathsf{Tr}_1 = (\sqrt{3}/2, 1/2)$, $\mathsf{Tr}_2 = (0, 1)$. Tr_i acts trivially on both the electromagnetic field and the polarization. Enforcing these symmetries, from Eqs. (S35) and (S36), we find

$$\tilde{\chi}^{MD}_{ijk} = \left[\begin{pmatrix} 0 & \tilde{\chi}^{MD}_{xxy} & 0 \\ \tilde{\chi}^{MD}_{xyx} & 0 & \tilde{\chi}^{MD}_{xyz} \\ 0 & \tilde{\chi}^{MD}_{xzy} & 0 \end{pmatrix} \begin{pmatrix} \tilde{\chi}^{MD}_{yxx} & 0 & \tilde{\chi}^{MD}_{yxz} \\ 0 & \tilde{\chi}^{MD}_{yyy} & 0 \\ \tilde{\chi}^{MD}_{yzx} & 0 & \tilde{\chi}^{MD}_{yzz} \end{pmatrix} \begin{pmatrix} 0 & \tilde{\chi}^{MD}_{zxy} & 0 \\ \tilde{\chi}^{MD}_{zyx} & 0 & \tilde{\chi}^{MD}_{zyz} \\ 0 & \tilde{\chi}^{MD}_{zzy} & 0 \end{pmatrix} \right]. \tag{S40}$$

Below T_c, the zigzag antiferromagnetic order is along the x-axis, which we define as l_x. It transforms as $l_x \xrightarrow{C_2'} -l_x$, $l_x \xrightarrow{\sigma_v} l_x$, $l_x \xrightarrow{\mathsf{Tr}_1} l_x$, and $l_x \xrightarrow{\mathsf{Tr}_2} -l_x$. Expressing

$$\tilde{\chi}^{MD}_{ijk} = \tilde{\chi}^{MD,(0)}_{ijk} + \tilde{\chi}^{MD,(1)}_{ijk} l_x + ... + \tilde{\chi}^{MD,(n)}_{ijk} (l_x)^n + ..., \tag{S41}$$

we find that $\tilde{\chi}^{MD,(1)}_{ijk} \equiv 0$, and the leading contribution from l_x to the SHG tensor is $\tilde{\chi}^{MD,(2)}_{ijk} l_x^2$. Furthermore, $\tilde{\chi}^{MD,(2)}_{ijk}$ maintains the same structural form as Eq. (S40). As a result, we conclude that the transition from the paramagnetic to the antiferromagnetic phase at T_c does not introduce any additional independent SHG tensor components for bulk $NiPS_3$.

The magnitude of the antiferromagnetic order parameter l_x across T_c can be determined by analyzing the SHG intensity at specific polarization angles.

For the XY configuration, using Eq. (S40), we find that when the wave vector is normal to the xy plane (i.e., $\theta = 0$),

$$\mathcal{A}_{\mathrm{SHG}}(\psi, XY) = \tilde{\chi}^{MD}_{xxy} \sin^3\psi - (\tilde{\chi}^{MD}_{xyx} + \tilde{\chi}^{MD}_{yxx} - \tilde{\chi}^{MD}_{yyy}) \sin\psi \cos^2\psi. \tag{S42}$$

At $\psi = 0$, $\mathcal{A}_{\mathrm{SHG}}(\psi = 0, XY) = 0$. At $\psi = \pi/2$, we find that $\mathcal{A}_{\mathrm{SHG}}(\psi = \pi/2, XY) = \chi_{xxy}$, and thus the SHG intensity across T_c is

$$I_{\mathrm{SHG}}(\psi = \pi/2, XY) = |\mathcal{A}_{\mathrm{SHG}}(\psi = \pi/2, XY)|^2 \tag{S43}$$

$$= |\tilde{\chi}^{MD,(0)}_{xxy}|^2 + 2l_x^2 \,\mathrm{Re}\left(\tilde{\chi}^{MD,(0)}_{xxy} (\tilde{\chi}^{MD,(2)}_{xxy})^*\right) + O(l_x^4). \tag{S44}$$

Similarly, for the XX configuration, we obtain $I_{\mathrm{SHG}}(\psi = \pi/2, XX) = 0$ and

$$I_{\mathrm{SHG}}(\psi = 0, XX) = |\mathcal{A}_{\mathrm{SHG}}(\psi = 0, XX)|^2 = |\tilde{\chi}^{MD,(0)}_{xyx}|^2 + 2l_x^2 \,\mathrm{Re}\left[\tilde{\chi}^{MD,(0)}_{xyx} (\tilde{\chi}^{MD,(2)}_{xyx})^*\right] + O(l_x^4). \tag{S45}$$

Monolayer: First, neglecting strain, we note that above T_c, the system retains the symmetries previously discussed for multilayer $NiPS_3$, with the addition of C_{3z} symmetry.

Enforcing C_{3z}, the SHG tensor above T_c takes the form

$$\tilde{\chi}^{MD}_{ijk} = \left[\begin{pmatrix} 0 & \tilde{\chi}^{MD}_{xxy} & 0 \\ \tilde{\chi}^{MD}_{xxy} & 0 & \tilde{\chi}^{MD}_{xyz} \\ 0 & \tilde{\chi}^{MD}_{xzy} & 0 \end{pmatrix} \begin{pmatrix} \tilde{\chi}^{MD}_{xxy} & 0 & -\tilde{\chi}^{MD}_{xyz} \\ 0 & -\tilde{\chi}^{MD}_{xxy} & 0 \\ \tilde{\chi}^{MD}_{xzy} & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & \tilde{\chi}^{MD}_{zxy} & 0 \\ -\tilde{\chi}^{MD}_{zxy} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right]. \tag{S46}$$

Below T_c, the zigzag order allows for six equivalent configurations, which transform as a vector. Defining the zigzag order as $\hat{q}l$, where $\hat{q} = [\cos(n\psi), \sin(n\psi)]$ with $n \in \mathbb{Z}_6$, $\psi = \pi/3$, the SHG tensor can be expressed as

$$\tilde{\chi}^{MD}_{ijk} = \tilde{\chi}^{MD,(0)}_{ijk} + l\tilde{\chi}^{MD,(1)}_{a,ijk} \hat{q}_a + l^2 \tilde{\chi}^{MD,(2)}_{ab,ijk} \hat{q}_a \hat{q}_b + ... \tag{S47}$$

We find that $\tilde{\chi}^{MD,(1)}_{a,ijk} \equiv 0$, and the leading order contribution from l to the SHG tensor is $l^2\tilde{\chi}^{MD,(2)}_{ab,ijk}\hat{q}_a\hat{q}_b$. Without loss of generality, considering the zigzag order along the x-axis, i.e., $\hat{q} = (1, 0)$, the SHG tensor takes the form

$$\tilde{\chi}^{MD,(2)}_{11,ijk} = \left[\begin{pmatrix} 0 & \tilde{\chi}^{MD,(2)}_{xxy} & 0 \\ \tilde{\chi}^{MD,(2)}_{xyx} & 0 & \tilde{\chi}^{MD,(2)}_{xyz} \\ 0 & \tilde{\chi}^{MD,(2)}_{xzy} & 0 \end{pmatrix}, \begin{pmatrix} \tilde{\chi}^{MD,(2)}_{yxx} & 0 & \tilde{\chi}^{MD,(2)}_{yxz} \\ 0 & \tilde{\chi}^{MD,(2)}_{yyy} & 0 \\ \tilde{\chi}^{MD,(2)}_{yzx} & 0 & \tilde{\chi}^{MD,(2)}_{yzz} \end{pmatrix}, \begin{pmatrix} 0 & \tilde{\chi}^{MD,(2)}_{zxy} & 0 \\ \tilde{\chi}^{MD,(2)}_{zyx} & 0 & \tilde{\chi}^{MD,(2)}_{zyz} \\ 0 & \tilde{\chi}^{MD,(2)}_{zzy} & 0 \end{pmatrix} \right]. \tag{S48}$$

Above T_c, the SHG intensity for the XY and XX configurations at normal incidence is

$$I_{\text{SHG}}(\psi, XY) = \frac{1}{2}|\tilde{\chi}^{MD,(0)}_{xxy}|^2 \left[1 - \cos(6\psi)\right], I_{\text{SHG}}(\psi, XX) = \frac{1}{2}|\tilde{\chi}^{MD,(0)}_{xxy}|^2 \left[1 + \cos(6\psi)\right]. \tag{S49}$$

Below T_N, the zigzag order breaks the C_{3z} symmetry, and the SHG polarimetry at l^2 order is described by

$$I^{(2)}_{\text{SHG}}(\psi, XY) = \frac{1}{2}\left[\cos(2\psi) - \cos(4\psi)\right] \text{Re}\left[(\tilde{\chi}^{MD,(0)}_{xxy})^*(\tilde{\chi}^{(MD,2)}_A + \tilde{\chi}^{(MD,2)}_B \cos(2\psi))\right], \tag{S50}$$

$$I^{(2)}_{\text{SHG}}(\psi, XX) = \frac{1}{2}\left[\cos(2\psi) + \cos(4\psi)\right] \text{Re}\left[(\tilde{\chi}^{MD,(0)}_{xxy})^*(-\tilde{\chi}^{(MD,2)}_A + \tilde{\chi}^{(MD,2)}_B \cos(2\psi))\right], \tag{S51}$$

where

$$\tilde{\chi}^{(MD,2)}_A = \left(\tilde{\chi}^{MD,(2)}_{xxy} - \tilde{\chi}^{MD,(2)}_{xyx} + \tilde{\chi}^{MD,(2)}_{yxx} - \tilde{\chi}^{MD,(2)}_{yyy}\right), \tag{S52}$$

$$\tilde{\chi}^{(MD,2)}_B = \left(\tilde{\chi}^{MD,(2)}_{xxy} + \tilde{\chi}^{MD,(2)}_{xyx} + \tilde{\chi}^{MD,(2)}_{yxx} - \tilde{\chi}^{MD,(2)}_{yyy}\right), \tag{S53}$$

and $\psi = \pi/2$ ($\psi = 0$) remains to be at the extrema of $I_{\text{SHG}}(\psi)$ for the XY (XX) configuration. The MD SHG intensity across T_c for the XY (XX) configuration at $\psi = \pi/2$ ($\psi = 0$) is

$$I_{\text{SHG}}(\psi = \pi/2, XY) = |\tilde{\chi}^{MD,(0)}_{xxy}|^2 + 2\,\text{Re}\left[\left(\tilde{\chi}^{MD,(0)}_{xxy}\right)^* \tilde{\chi}^{MD,(2)}_{xyx}\right] l^2 + O(l^4), \tag{S54}$$

$$I_{\text{SHG}}(\psi = 0, XX) = |\tilde{\chi}^{MD,(0)}_{xxy}|^2 + 2\,\text{Re}\left[\left(\tilde{\chi}^{MD,(0)}_{xxy}\right)^* \tilde{\chi}^{MD,(2)}_{xyx}\right] l^2 + O(l^4). \tag{S55}$$

To summarize, from Eqs. (S42) and (S44) for multilayer, and Eqs. (S54) and (S55) for monolayer, we find that the SHG intensity for the XY configuration at $\psi = \pi/2$ (and for the XX configuration at $\psi = 0$) is expressed similarly for bulk and monolayer $NiPS_3$.

Supplementary Tables and Figures

Sample	$\lvert\tilde{\chi}_0\rvert$ (arb. u.)	χ_2^{MD} (arb. u.)	β	T_c (K)
1L	1.357 ± 0.005	0.41 ± 0.08	0.24 ± 0.04	141.1 ± 0.9
2L	2.259 ± 0.02	1.46 ± 0.05	0.31 ± 0.02	140.0 ± 0.9
3L	2.007 ± 0.04	1.91 ± 0.1	0.33 ± 0.03	148.1 ± 2.1
4L	1.712 ± 0.03	1.56 ± 0.08	0.30 ± 0.03	148.3 ± 1.9

Table S1 | Fitting parameters for the temperature-dependent SHG data. The fits correspond to those shown in Fig. S17 and underlie the results reported in the main text. Data were fitted to the model described in Eq. (S8) using a nonlinear least-squares routine in which all parameters were allowed to vary freely. The quoted uncertainties correspond to 95% confidence intervals obtained from the covariance matrix of the fit. For the monolayer sample, only data acquired at temperatures at or above the transition temperature of 120 K were included in the fits. For the remaining thicknesses, all data points were included.

Sample	$\lvert\tilde{\chi}_0\rvert$ (arb. u.)	χ_2^{MD} (arb. u.)	β	T_c (K)
1L	1.358 ± 0.004	0.40 ± 0.09	0.23 ± 0.05	140.4 ± 1.4
2L	2.259 ± 0.010	1.47 ± 0.05	0.31 ± 0.02	140.1 ± 1.7
3L	2.011 ± 0.017	2.05 ± 0.21	0.35 ± 0.05	147.4 ± 3.3
4L	1.706 ± 0.042	1.55 ± 0.08	0.30 ± 0.03	148.3 ± 1.5

Table S2 | Bootstrap-derived fitting parameters for the temperature-dependent SHG data. A bootstrap resampling procedure was performed in which randomly drawn datasets (with replacement) were repeatedly fitted to the model in Eq. (S8) using a nonlinear least-squares routine. In each fit, all model parameters were allowed to vary freely. The tabulated values correspond to the mean of 10^4 such fits, and the quoted uncertainties denote 95% confidence intervals obtained from the resulting parameter distributions using a bias-corrected percentile method. For the monolayer sample, only data acquired at temperatures at or above the transition temperature of 120 K were included in the bootstrap routine. For the remaining thicknesses, all data points were included.

Sample	$\lvert\tilde{\chi}_0\rvert$ (arb. u.)	χ_2^{MD} (arb. u.)	β	T_c (K)
1L	1.43	0.45	0.23 ± 0.04	140.5 ± 0.9
2L	2.30	1.44	0.32 ± 0.02	141.0 ± 0.9
3L	2.07	1.87	0.33 ± 0.03	147.8 ± 1.9
4L	1.78	1.54	0.31 ± 0.03	148.1 ± 1.7

Table S3 | Alternative Fitting parameters for temperature-dependent SHG data. The data were fitted using the model described in Eq. (S8). A nonlinear least-squares optimization procedure was performed, with the parameter $\lvert\tilde{\chi}_0\rvert$ fixed to the average value of the high-temperature SHG signal. Quoted uncertainties correspond to 95% confidence intervals. For the monolayer sample, only data acquired at temperatures at or above the transition temperature of 120 K were included in the fits. For the remaining thicknesses, all data points were included.

Sample	T_c (K)
1L	–
2L	142 ± 5
3L	148.3 ± 0.7
4L	146 ± 8

Table S4 | Transition temperatures extracted from Raman data. The transition temperatures are extracted from fits to the temperature dependence of the P_2 peak splitting using the same functional form as in Eq. (S8). The layer dependence mirrors that observed in the SHG measurements; however, sparse sampling near the critical region limits the accuracy of the extracted transition temperatures. Quoted uncertainties correspond to 95% confidence intervals.

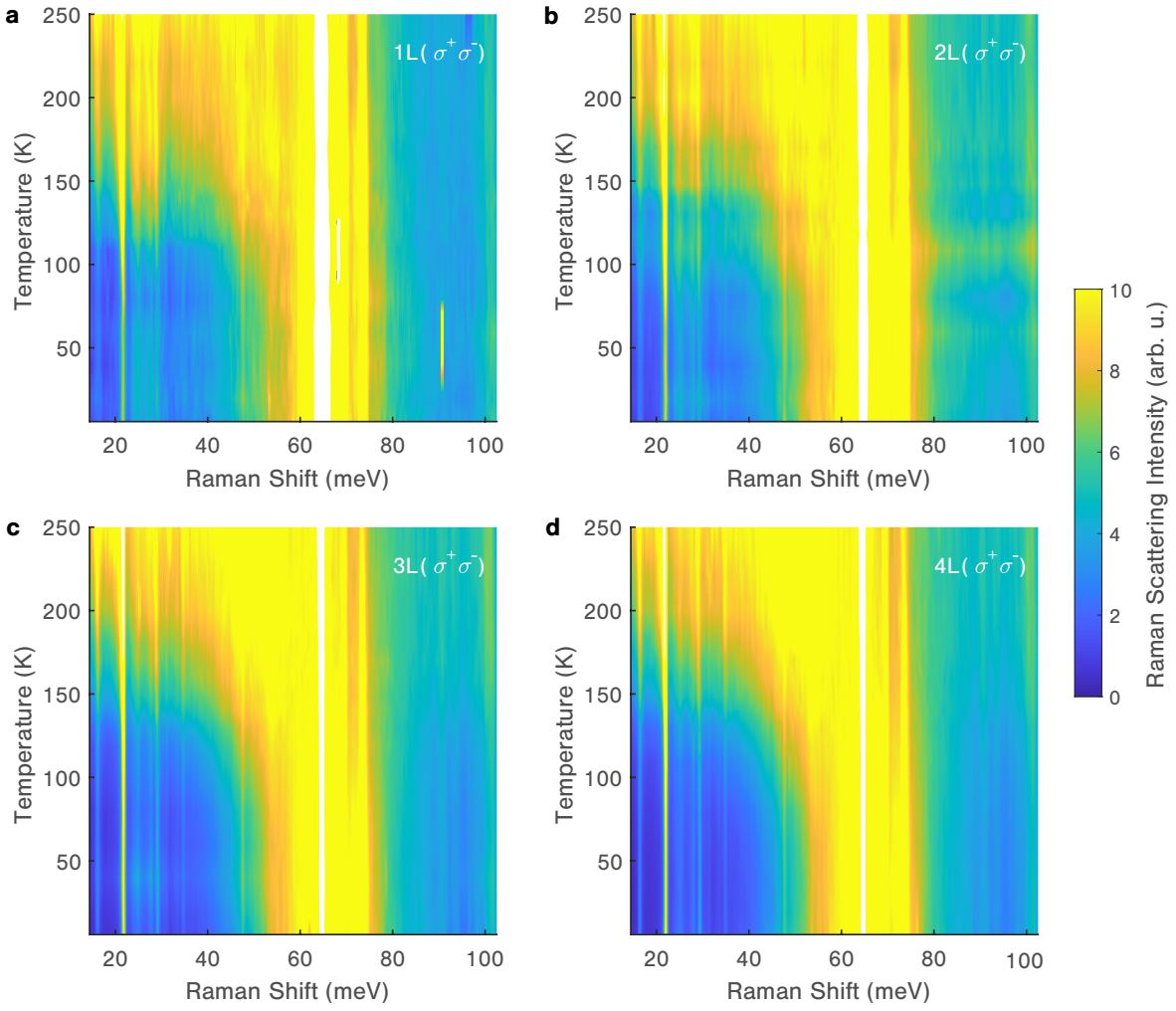


Fig. S1 | Temperature-dependent Raman spectra of $NiPS_3$ thin layers. Temperature-dependent Raman spectra in the cross-circularly polarized $\sigma^+\sigma^-$ configuration for **a**, monolayer, **b**, bilayer, **c**, trilayer, and **d**, tetralayer $NiPS_3$. The white peak at 65 meV corresponds to a phonon mode of the silicon substrate, while the broad yellow feature is the bimagnon continuum. Across all thicknesses, the bimagnon feature emerges around 68 meV at low temperatures, gradually broadening and redshifting as the temperature increases.

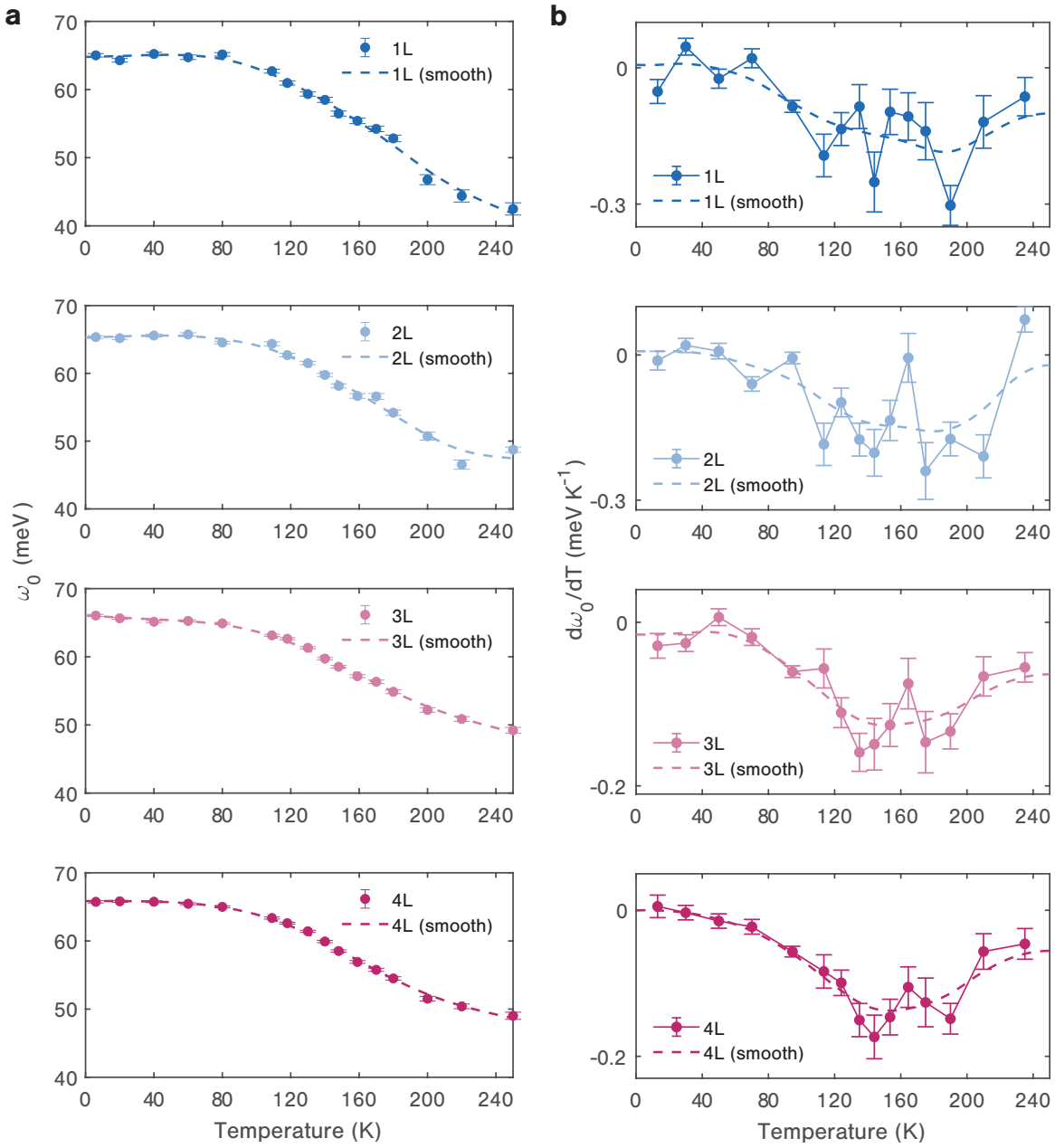


Fig. S2 | Fit parameters for temperature-dependent bimagnon Raman scattering. **a**, Temperature dependence of the bimagnon peak position, ω_0, extracted from the spectra shown in Fig. S1. **b**, Derivative of bimagnon peak energy with respect to temperature, $d\omega_0/dT$. An extremum in $d\omega_0/dT$ appears in the range of 140 K to 160 K, in agreement with Ref. [3]. While this feature could suggest the presence of a magnetic phase transition in monolayer $NiPS_3$, it does not provide a definitive signature of the system's critical scaling behavior. Error bars correspond to 95% confidence intervals obtained from the fits.

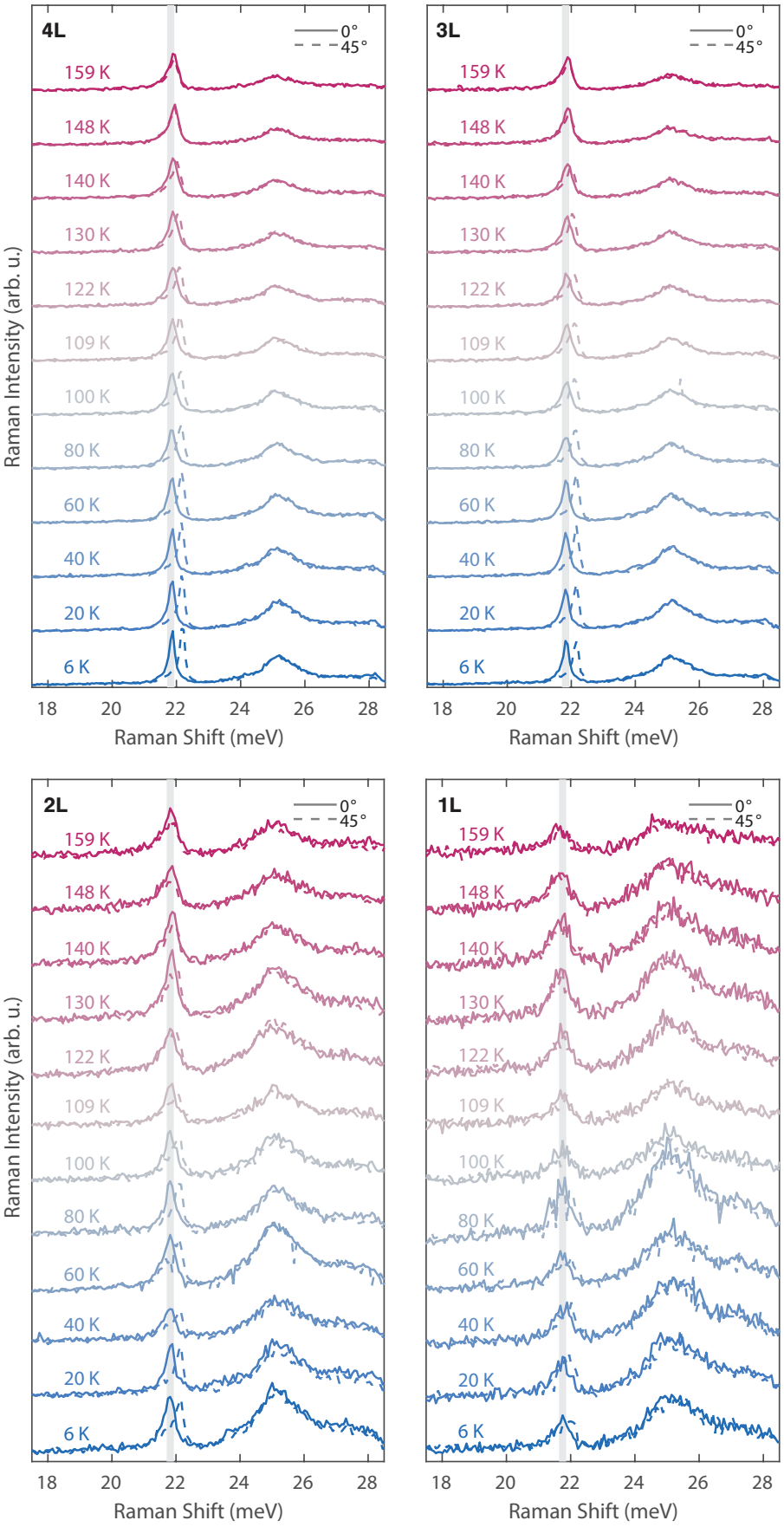


Fig. S3 | Raw temperature-dependent Raman spectra around P_2 splitting in $NiPS_3$ thin layers. Temperature-dependent spontaneous Raman spectra of 4L, 3L, 2L, and 1L $NiPS_3$, revealing the progressive splitting of the P_2 phonon mode. Data are shown for two incident polarization angles (0° and 45°) to highlight polarization-dependent features. No smoothing of the curves was performed.

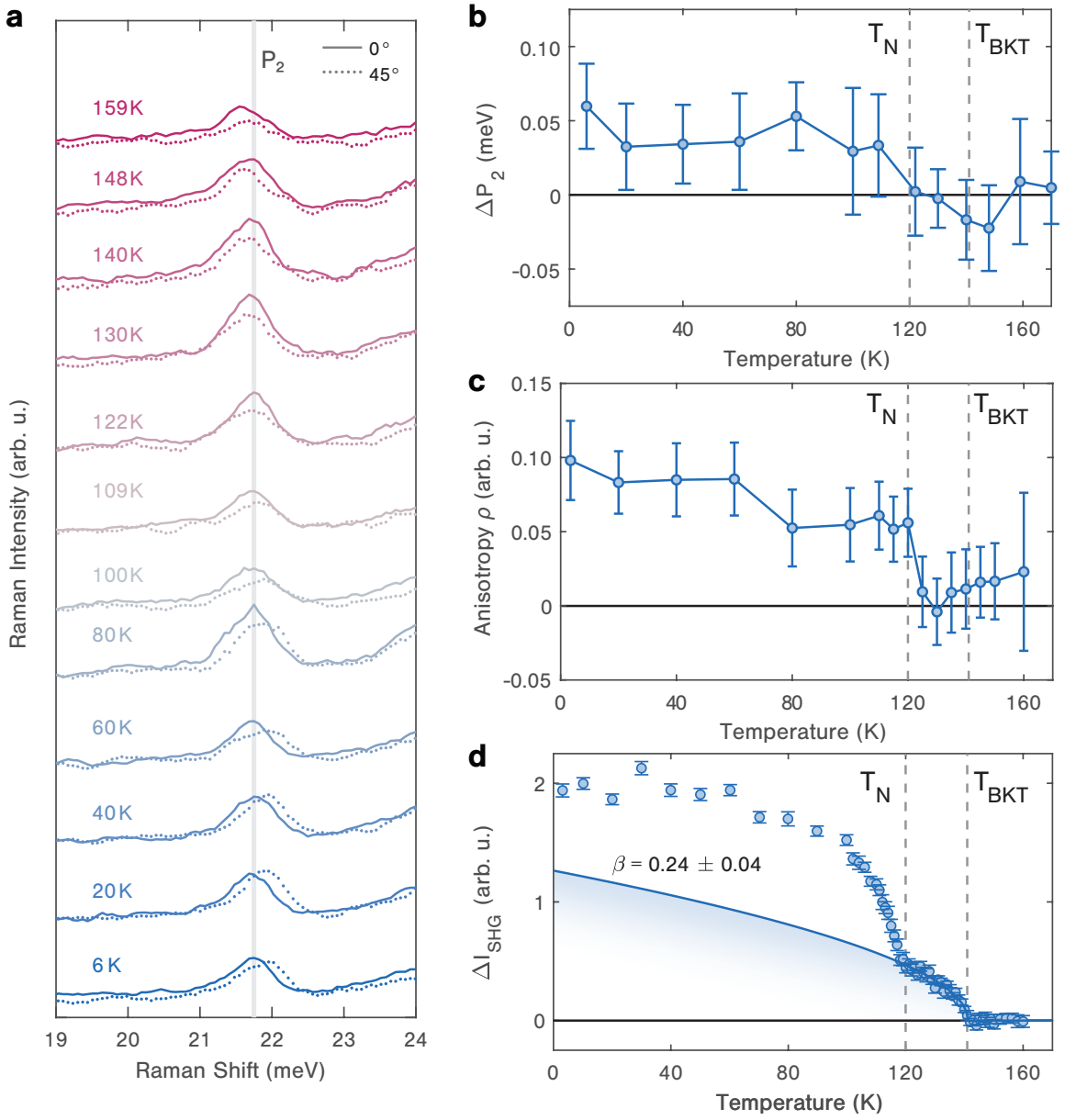


Fig. S4 | Raman and SHG characterization of magnetic order in monolayer $NiPS_3$. a, Temperature-dependent spontaneous Raman scattering spectra on monolayer $NiPS_3$ showing the splitting of the P_2 phonon mode for two incident polarization configurations (0° and 45°). The Raman data shown have been smoothed for clarity. **b,** Temperature-dependent P_2 phonon mode splitting energy (ΔP_2) derived from fits to the raw unsmoothed data shown in Fig. S3. The plot is reproduced from Fig. 2e and is shown with a magnified vertical scale to emphasize the signal. A finite ΔP_2 is observed at low temperatures and disappears around T_N = 120 K. Raman peaks were fitted to the raw (unsmoothed) data using a Lorentzian function, constrained to yield a conservative estimate of peak splitting across all wavelengths. Unconstrained fits yield the same exact trend, with the same transition temperature, but with low-temperature splittings larger by a factor of 2-3. **c**, Temperature-dependent SHG anisotropy, and **d**, SHG intensity change for the same monolayer sample, reproduced from Figs. 4c and 3d, respectively. Both transition temperatures (T_N = 120 K and T_{BKT} = 141 K) are indicated. Error bars are defined in the main text.

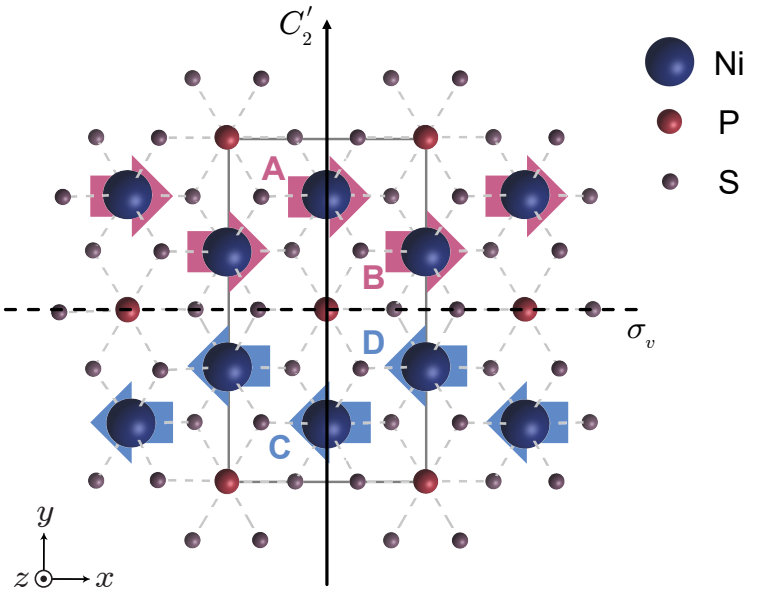


Fig. S5 | Magnetic unit cell of multilayer $NiPS_3$. Schematic of the 4-sublattice magnetic unit cell of multilayer $NiPS_3$, illustrating the magnetic sublattice sites (m_A, m_B, m_C, m_D). Also depicted are the magnetocrystalline symmetries of bulk monoclinic $NiPS_3$, which are used for the symmetry analysis in Supplementary Note 6. These include a vertical mirror plane (σ_v) and an in-plane two-fold rotation axis (C_2') along the y-axis.

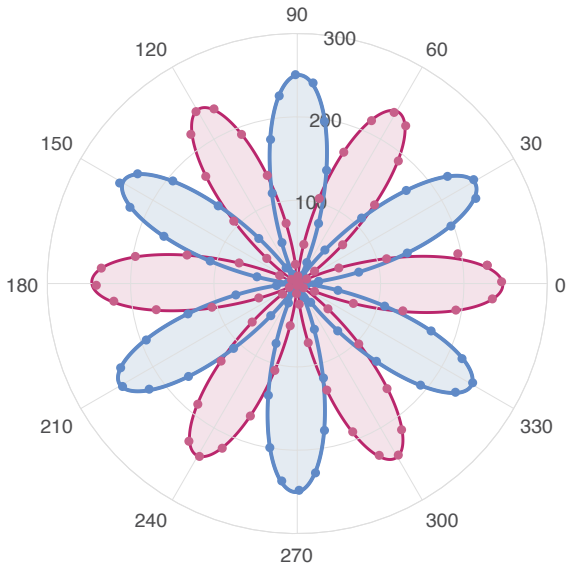


Fig. S6 | SHG polarimetry on GaAs (111). Reference anisotropic SHG patterns measured on GaAs(111) exhibit undistorted six-fold symmetry in both the parallel- (*blue*) and cross-polarized (*red*) channels, serving as a benchmark for assessing SHG anisotropy in $NiPS_3$.

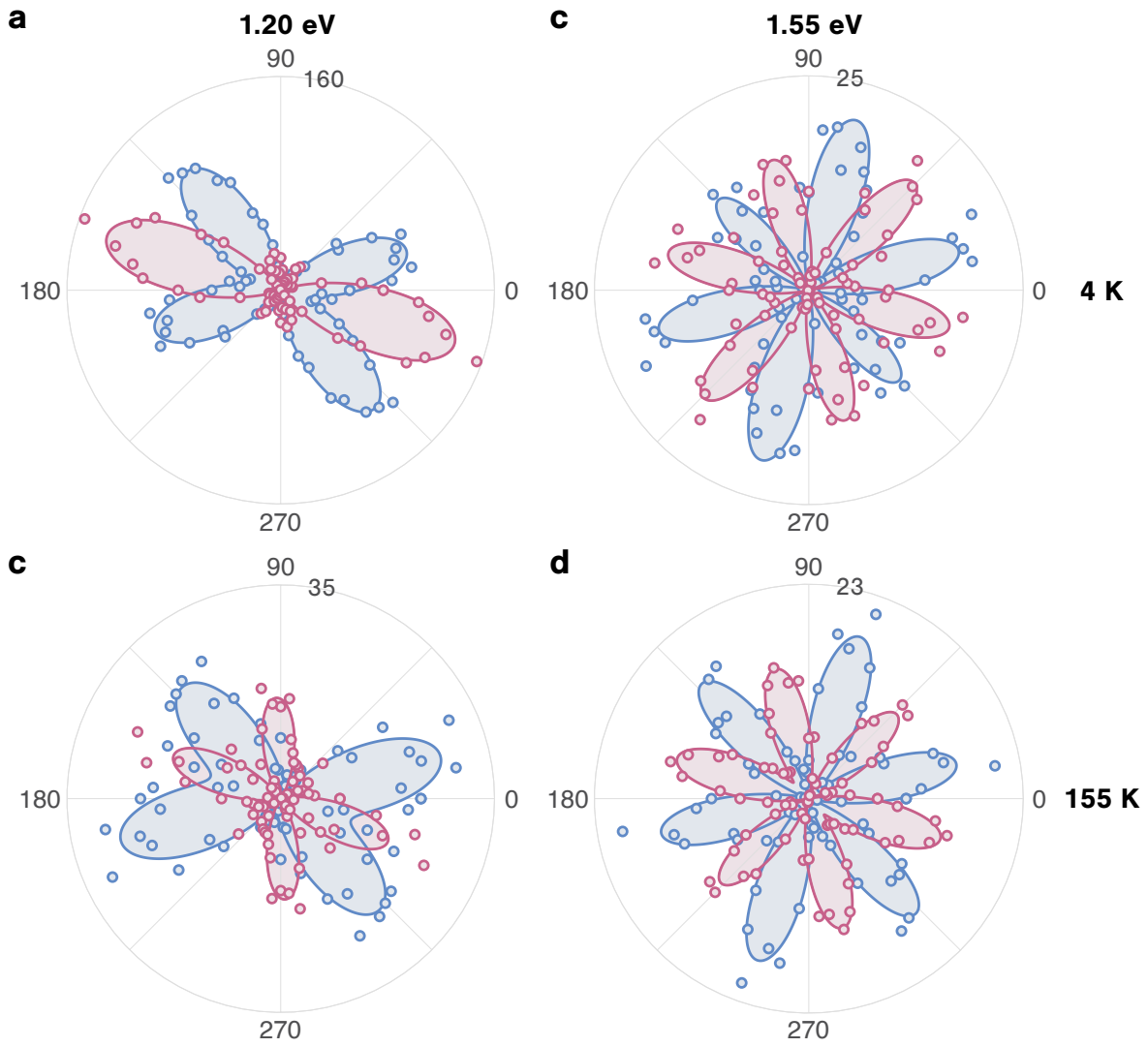


Fig. S7 | Probe photon energy-dependent SHG polarimetry patterns on $NiPS_3$. SHG polar patterns measured on a thin $NiPS_3$ flake using a probe photon energy of 1.20 eV are shown **a**, far below and **c**, near T_N. Similarly, SHG polar patterns using a 1.55 eV probe are displayed for the same sample **b**, far below and **d**, near T_N. Blue (red) circles represent experimental data from co- (cross-) linear polarization geometry, while solid lines indicate corresponding fits. At 1.55 eV, the SHG signal is quite weak and exhibits minimal variation across T_N, in agreement with previous reports[14]. This is expected, as the probe is off-resonant with nearby d-d transitions, resulting in a weak, relatively isotropic EQ/surface SHG response. In contrast, at 1.20 eV – the same probe photon energy used in the main text – SHG is more sensitive to MD due to resonance with d-d transitions. As a result, the low-temperature SHG signal is substantially stronger and shows a pronounced C_2 anisotropy. The presence of a C_2 distortion near T_N may originate from the intrinsic monoclinic stacking of the $NiPS_3$ lattice.

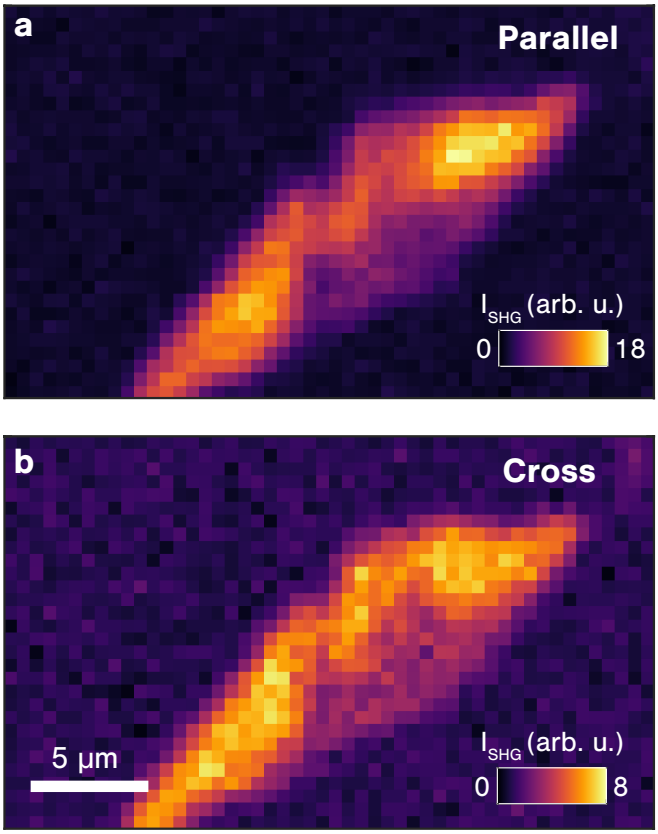


Fig. S8 | SHG imaging of few-layer $NiPS_3$ flake. Spatially resolved SHG maps of the multilayer flake (see Fig. 2a-b in the main text) measured at 3 K in both **a**, parallel- and **b**, cross-polarized detection configurations. The distinct SHG contrast enables clear identification of all four flake regions, ranging from monolayer to tetralayer.

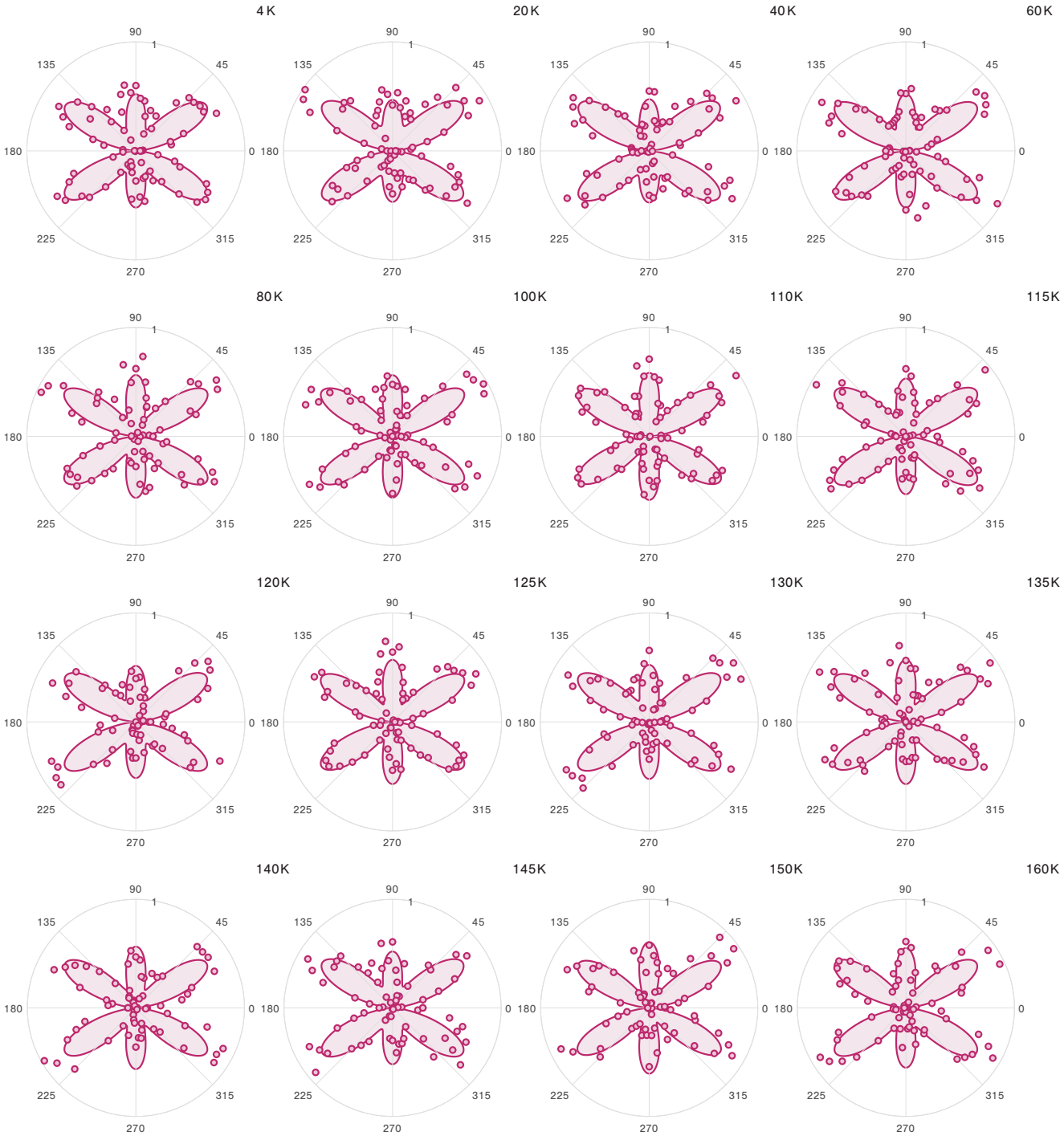


Fig. S9 | Temperature dependent SHG polarimetry patterns for monolayer $NiPS_3$. Raw polarimetry data corresponding to Fig. 4c, displaying the temperature-dependent anistropic SHG response in monolayer $NiPS_3$. Data are presented for the parallel-polarized detection channels, with experimental measurements (*circles*) shown alongside fitted curves (*lines*). SHG signals are normalized for comparison. Data are shown as a function of the in-plane polarization angle ϕ, measured from the crystallographic C_2' axis (the y-axis), where $\phi = \psi + \pi/2$.

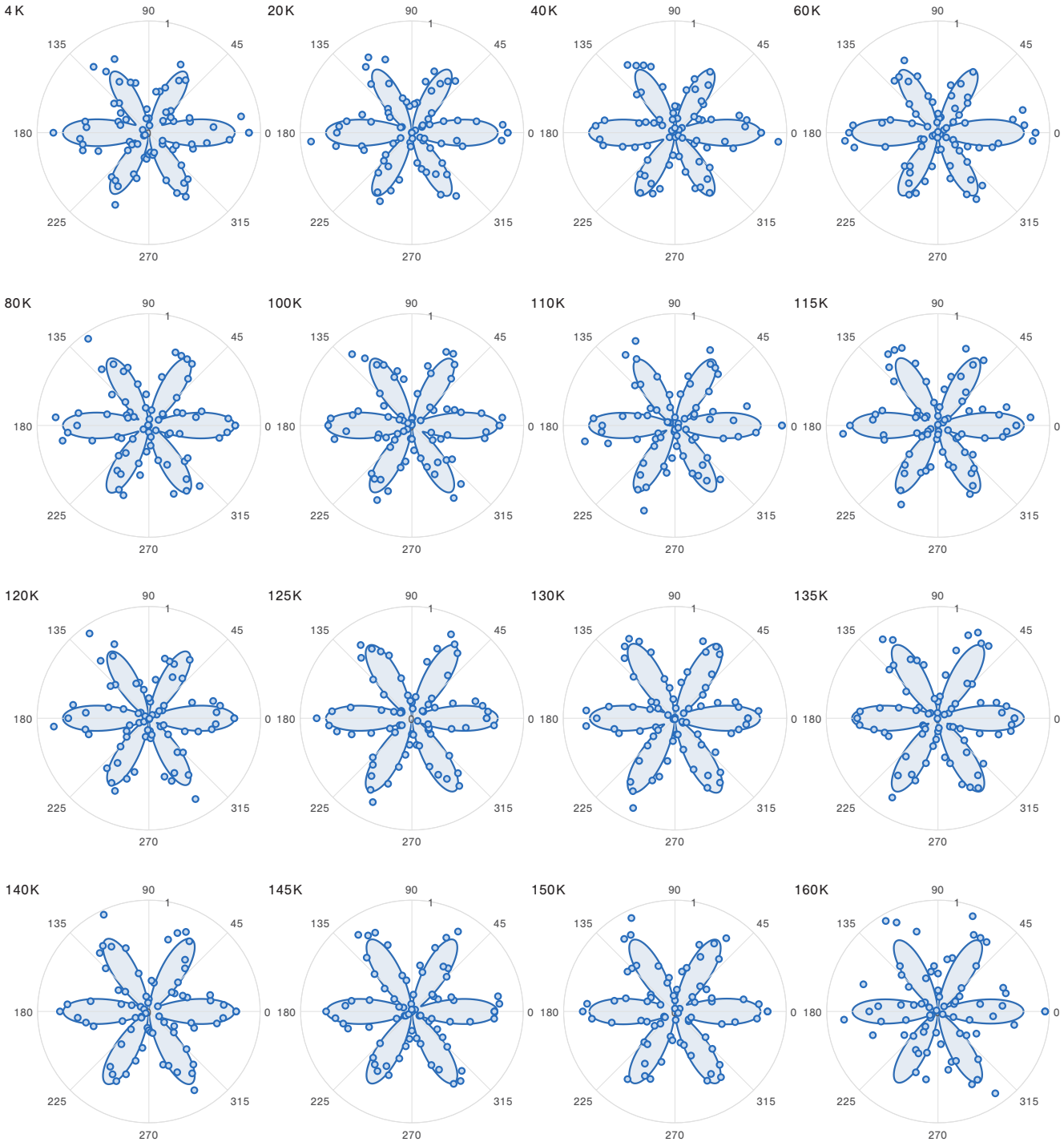


Fig. S10 | Temperature dependent SHG polarimetry patterns for monolayer $NiPS_3$. Raw polarimetry data corresponding to Fig. 4c, displaying the temperature-dependent anistropic SHG response in monolayer $NiPS_3$. Data are presented for the cross-polarized detection channels, with experimental measurements (*circles*) shown alongside fitted curves (*lines*). SHG signals are normalized for comparison. Data are shown as a function of the in-plane polarization angle ϕ, measured from the crystallographic C_2' axis (the y-axis), where $\phi = \psi + \pi/2$.

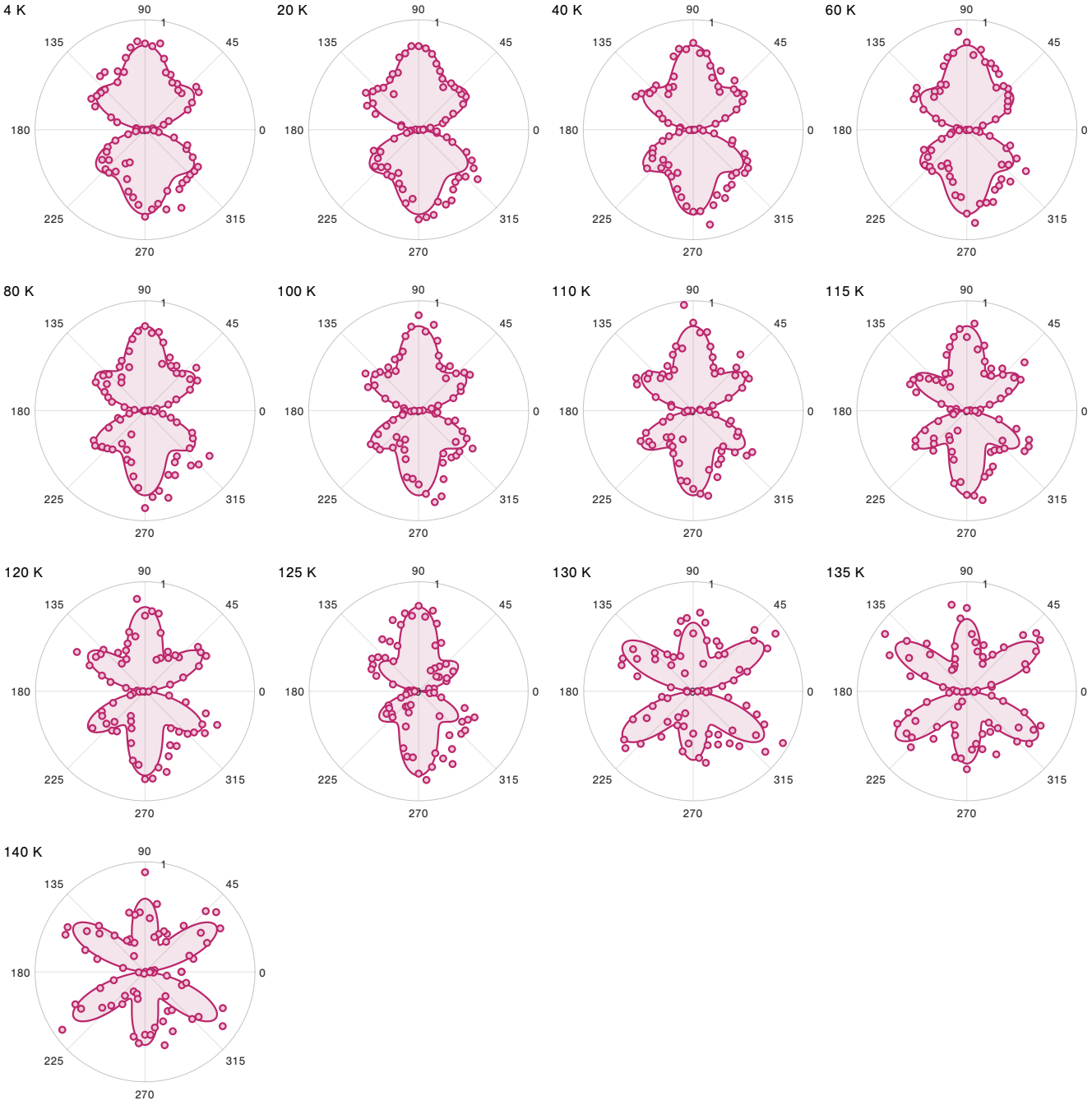


Fig. S11 | Temperature dependent SHG polarimetry patterns for bilayer $NiPS_3$. Raw polarimetry data corresponding to Fig. 4c, displaying the temperature-dependent anistropic SHG signal in bilayer $NiPS_3$. Data are presented for the parallel-polarized detection channels, with experimental measurements (*circles*) shown alongside fitted curves (*lines*). SHG signals are normalized for comparison. Data are shown as a function of the in-plane polarization angle ϕ, measured from the crystallographic C_2' axis (the y-axis), where $\phi = \psi + \pi/2$.

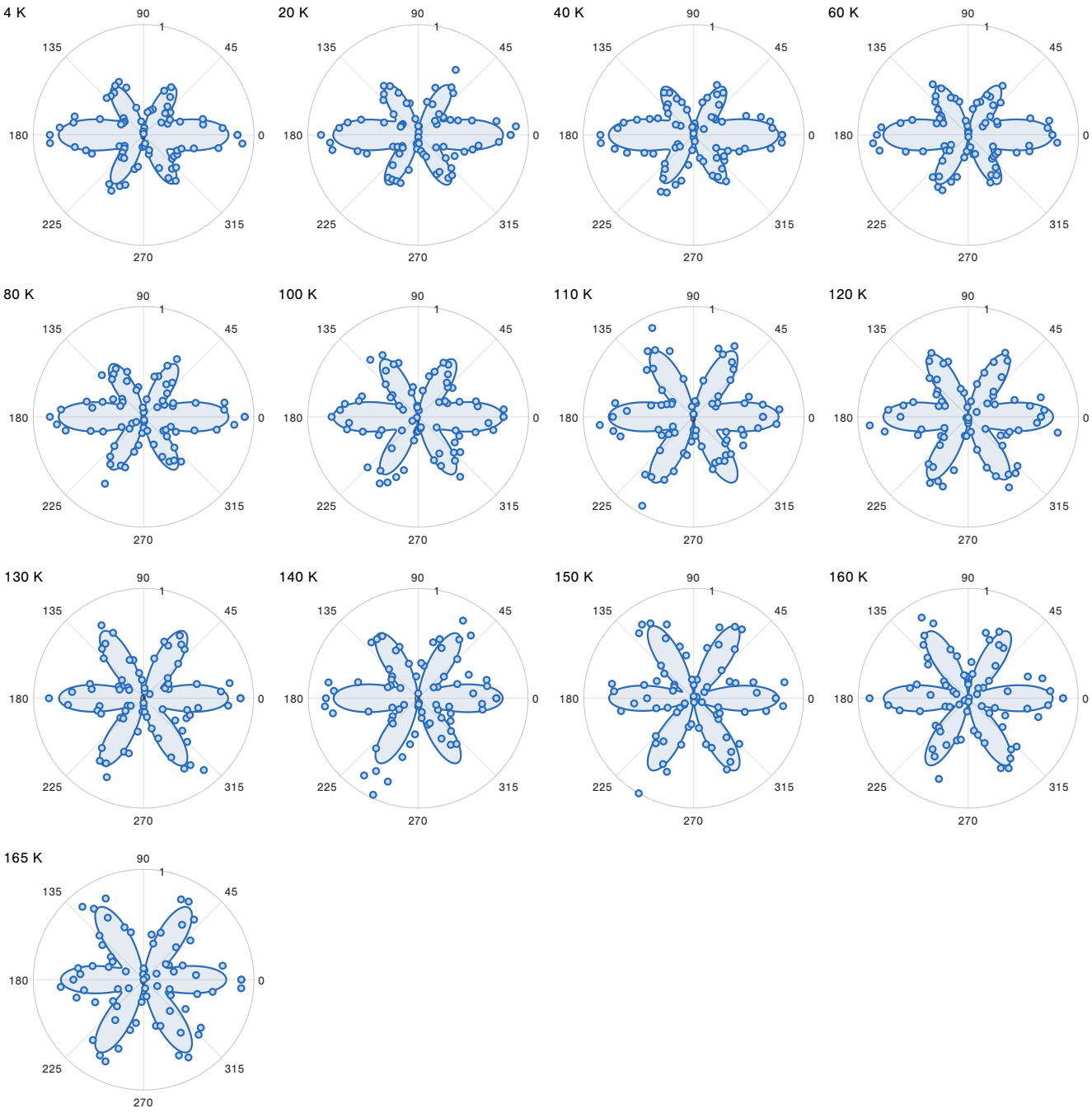


Fig. S12 | Temperature dependent SHG polarimetry patterns for bilayer $NiPS_3$. Raw polarimetry data corresponding to Fig. 4c, displaying the temperature-dependent anistropic SHG signal in bilayer $NiPS_3$. Data are presented for the cross-polarized detection channels, with experimental measurements (*circles*) shown alongside fitted curves (*lines*). SHG signals are normalized for comparison. Data are shown as a function of the in-plane polarization angle ϕ, measured from the crystallographic C_2' axis (the y-axis), where $\phi = \psi + \pi/2$.

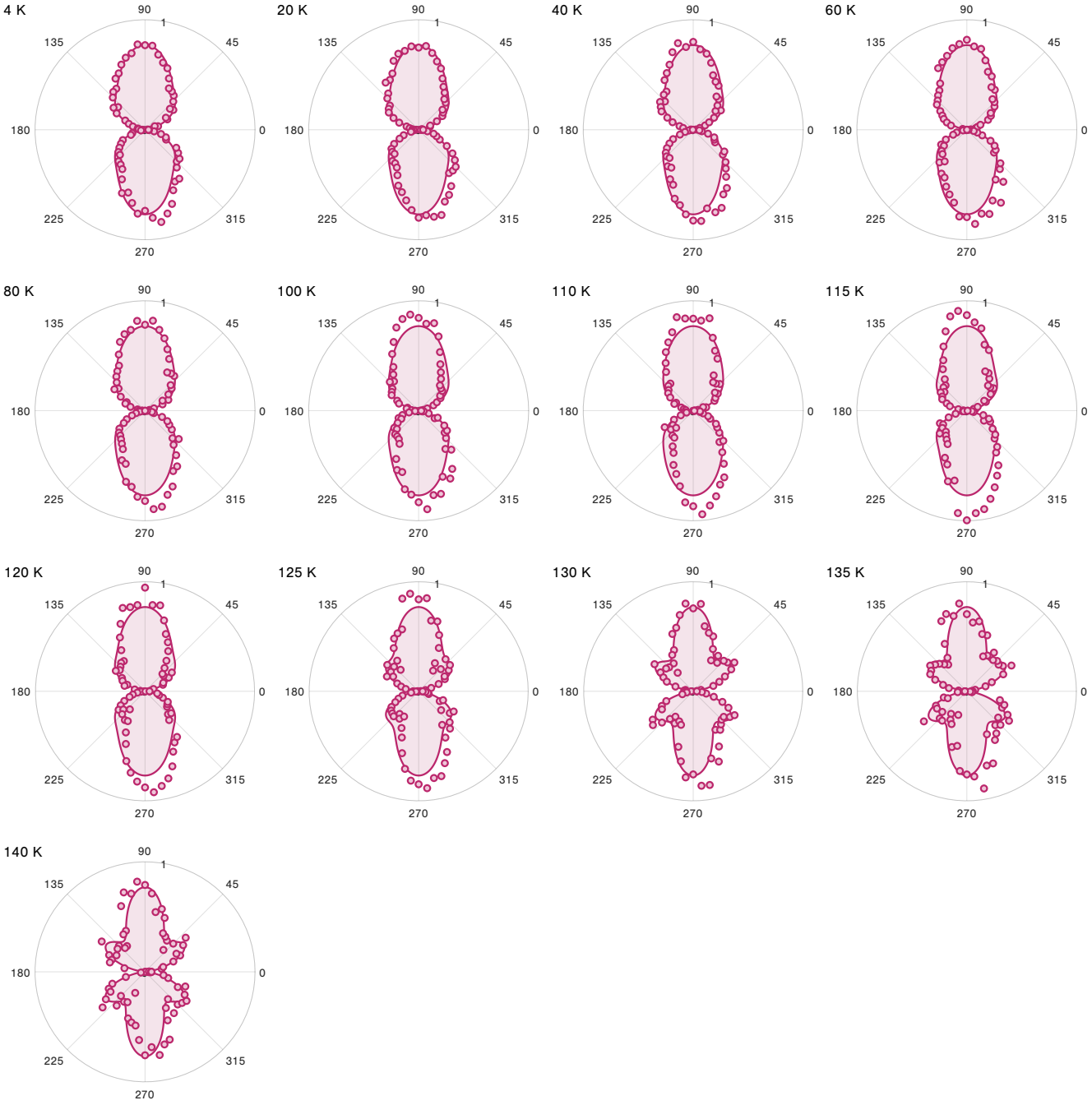


Fig. S13 | Temperature dependent SHG polarimetry patterns for trilayer $NiPS_3$. Raw polarimetry data corresponding to Fig. 4c, displaying the temperature-dependent anistropic SHG signal in trilayer $NiPS_3$. Data are presented for the parallel-polarized detection channels, with experimental measurements (*circles*) shown alongside fitted curves (*lines*). SHG signals are normalized for comparison. Data are shown as a function of the in-plane polarization angle ϕ, measured from the crystallographic C_2' axis (the y-axis), where $\phi = \psi + \pi/2$.

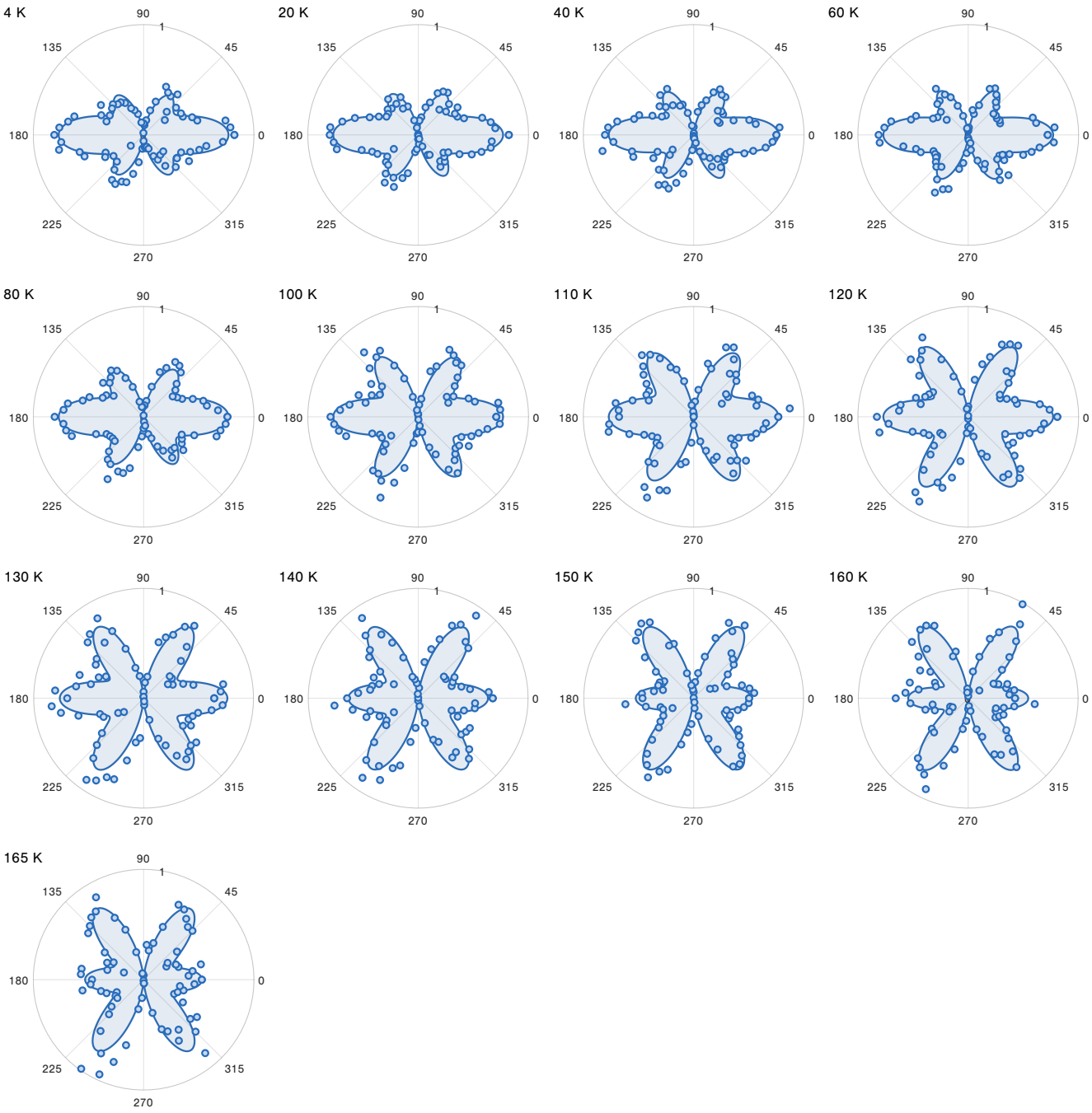


Fig. S14 | Temperature dependent SHG polarimetry patterns for trilayer $NiPS_3$. Raw polarimetry data corresponding to Fig. 4c, displaying the temperature-dependent anistropic SHG signal in trilayer $NiPS_3$. Data are presented for the cross-polarized detection channels, with experimental measurements (*circles*) shown alongside fitted curves (*lines*). SHG signals are normalized for comparison. Data are shown as a function of the in-plane polarization angle ϕ, measured from the crystallographic C_2' axis (the y-axis), where $\phi = \psi + \pi/2$.

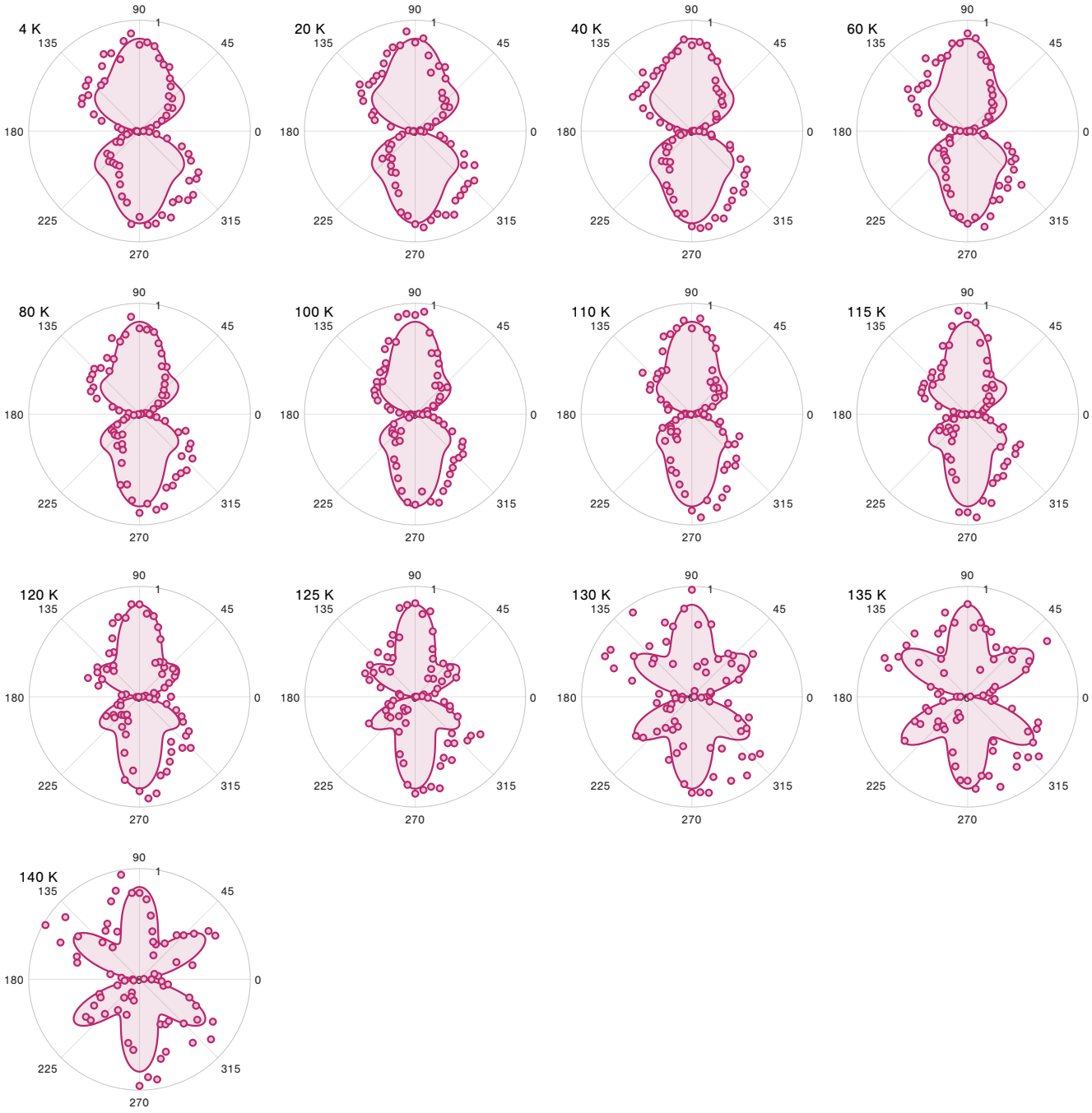


Fig. S15 | Temperature dependent SHG polarimetry patterns for tetralayer $NiPS_3$. Raw polarimetry data corresponding to Fig. 4c, displaying the temperature-dependent anistropic SHG signal in tetralayer $NiPS_3$. Data are presented for the parallel-polarized detection channels, with experimental measurements (*circles*) shown alongside fitted curves (*lines*). SHG signals are normalized for comparison. Data are shown as a function of the in-plane polarization angle ϕ, measured from the crystallographic C_2' axis (the y-axis), where $\phi = \psi + \pi/2$.

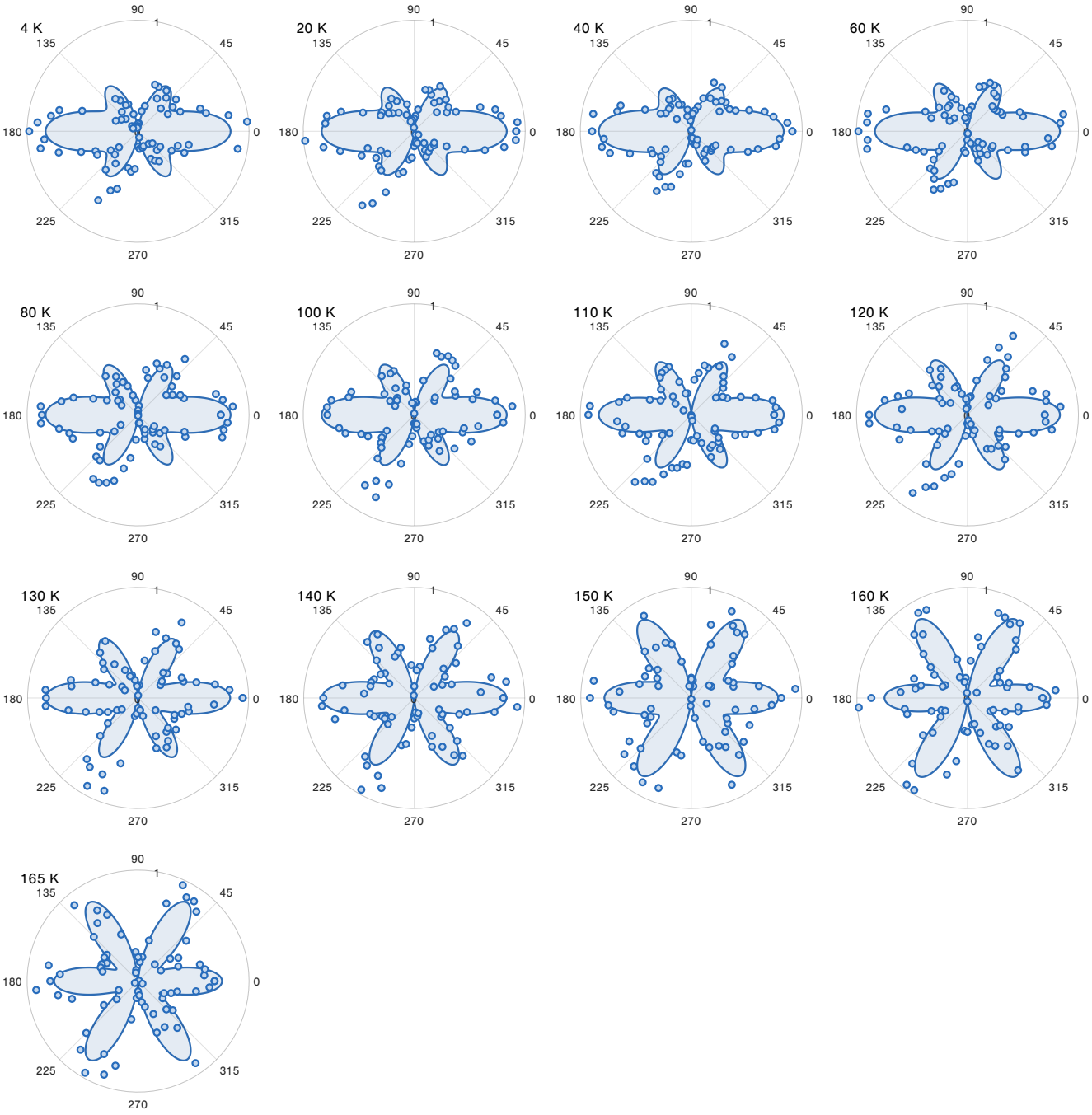


Fig. S16 | Temperature dependent SHG polarimetry patterns for tetralayer $NiPS_3$. Raw polarimetry data corresponding to Fig. 4c, displaying the temperature-dependent anistropic SHG signal in tetralayer $NiPS_3$. Data are presented for the cross-polarized detection channels, with experimental measurements (*circles*) shown alongside fitted curves (*lines*). SHG signals are normalized for comparison. Data are shown as a function of the in-plane polarization angle ϕ, measured from the crystallographic C_2' axis (the y-axis), where $\phi = \psi + \pi/2$.

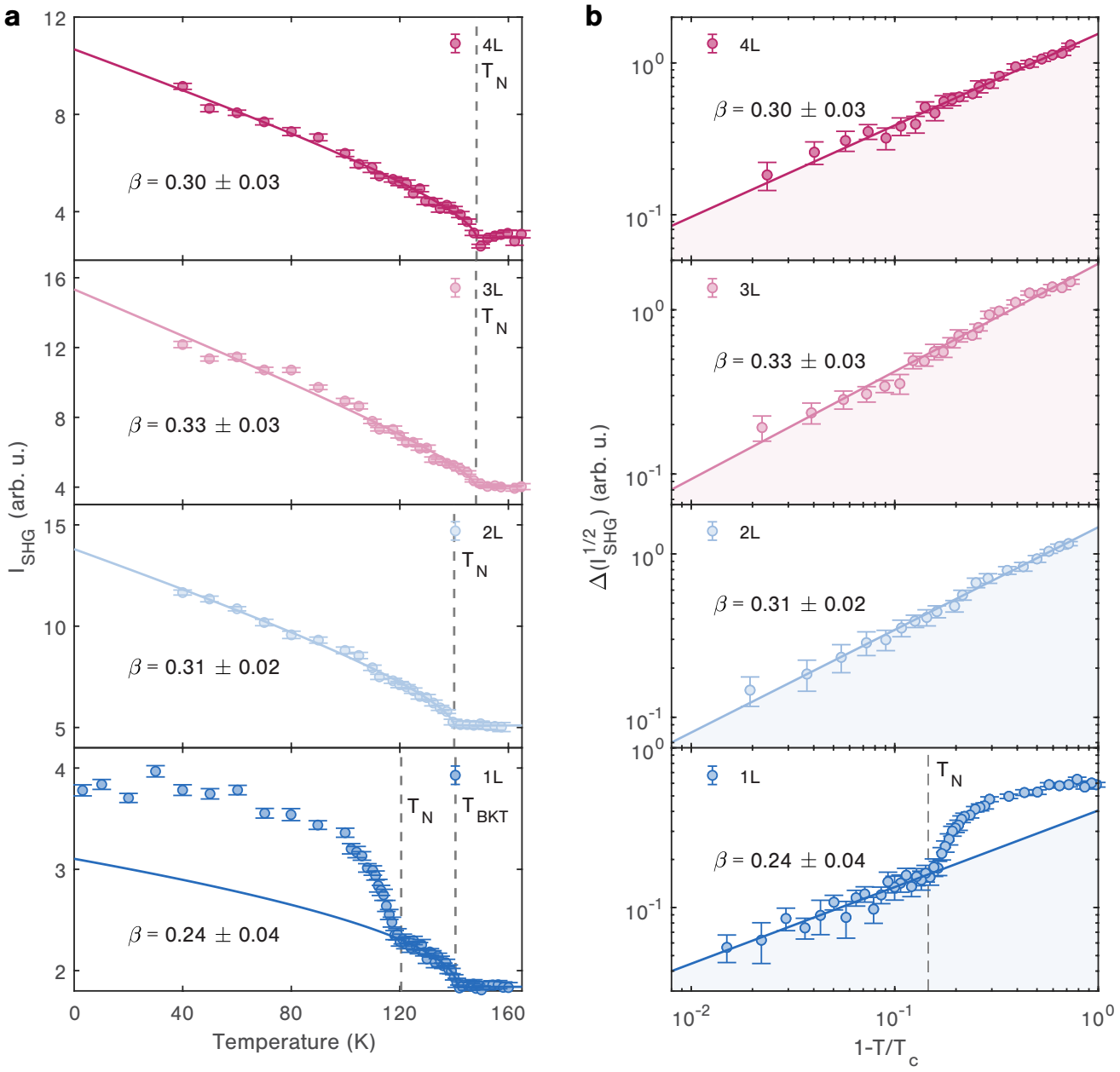


Fig. S17 | Complete temperature-dependent SHG signal. **a**, Raw SHG intensity as a function of temperature for samples of varying thickness, shown without subtraction of the high-temperature background. This dataset corresponds to that presented in Fig. 3c-d, prior to background subtraction. **b**, Log–log plot of the temperature-dependent SHG intensity for samples of varying thickness, also shown without subtraction of the high-temperature background. Extracted critical exponents β are indicated, with uncertainties given at the 95% confidence level. Error bars likewise represent 95% confidence intervals. All fit parameters are reported in Table S1. Data values, including the number of samples, are defined in the Methods.

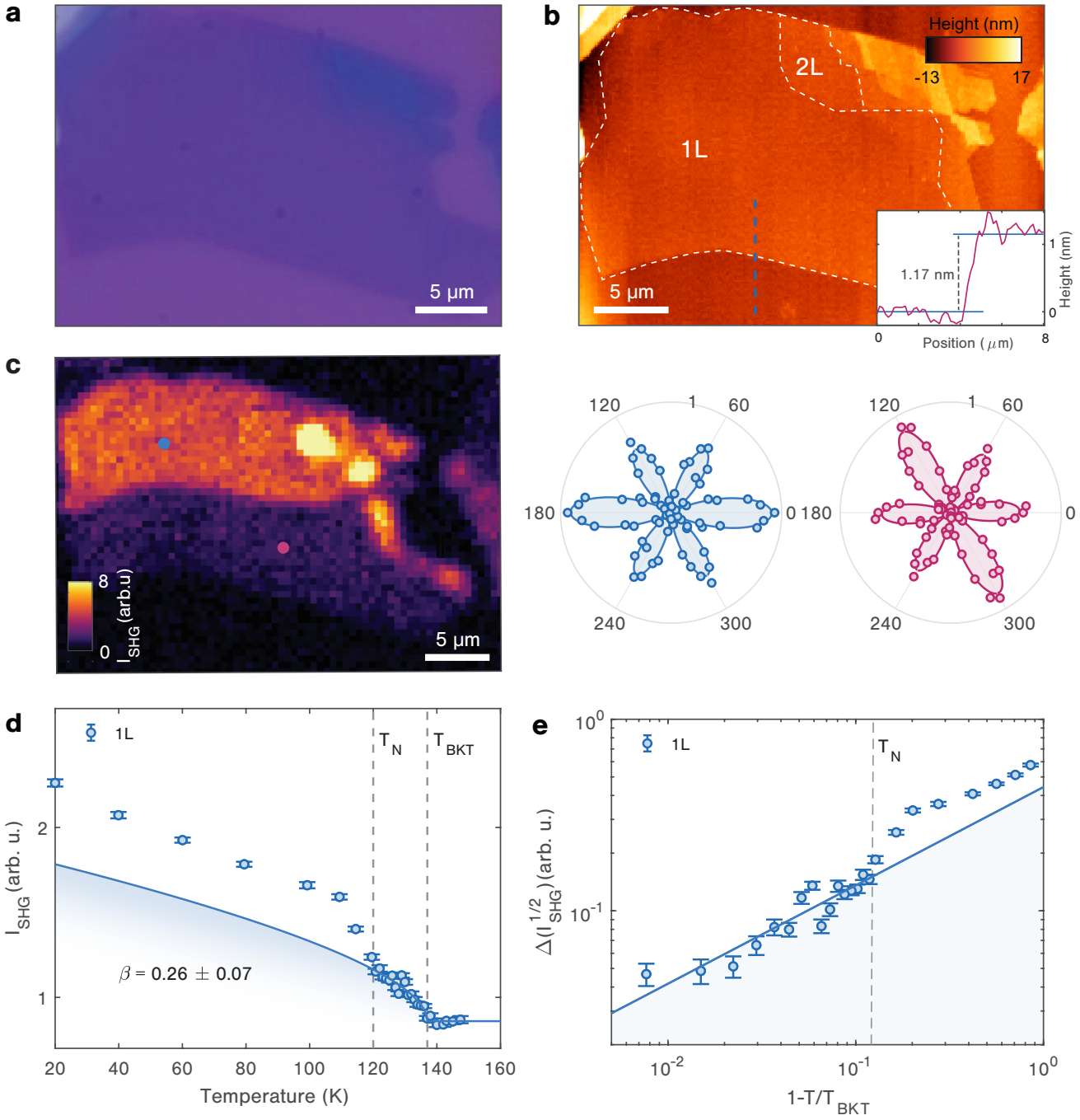


Fig. S18 | Critical scaling behavior of another monolayer $NiPS_3$ sample. **a**, Optical and **b**, atomic force microscope images of an additional monolayer $NiPS_3$ sample, distinct from the one presented in the main text. Inset of panel (b) depicts a height profile along the dashed blue line, confirming that the sample is a monolayer. **c**, (*left*) SHG spatial map of the same monolayer sample at 3 K, revealing two orientational domains; (*right*) corresponding SHG polar patterns extracted from the top (blue) and bottom (red) regions. Two representative locations within the regions are marked by blue and red dots. The patterns are rotated by $60°$, consistent with the distinct domain orientation of the long-range antiferromagnetic order. **d-e**, Temperature-dependent SHG intensity collected on the blue dot above, plotted on linear and log-log axes. Note that T_{BKT} = 138 K. Solid lines indicate fits to an order parameter model (Eq. 2 in the main text). Below T_{BKT}, the data yield a critical exponent of $\beta = 0.26 \pm 0.07$ (95% confidence intervals), consistent with 2D XY universality. A pronounced kink at $T_N = 120$ K marks a second transition into a long-range ordered phase, as expected from the six-state clock model. SHG data are an average of $N = 42$ observations. Error bars represent 95% confidence intervals.

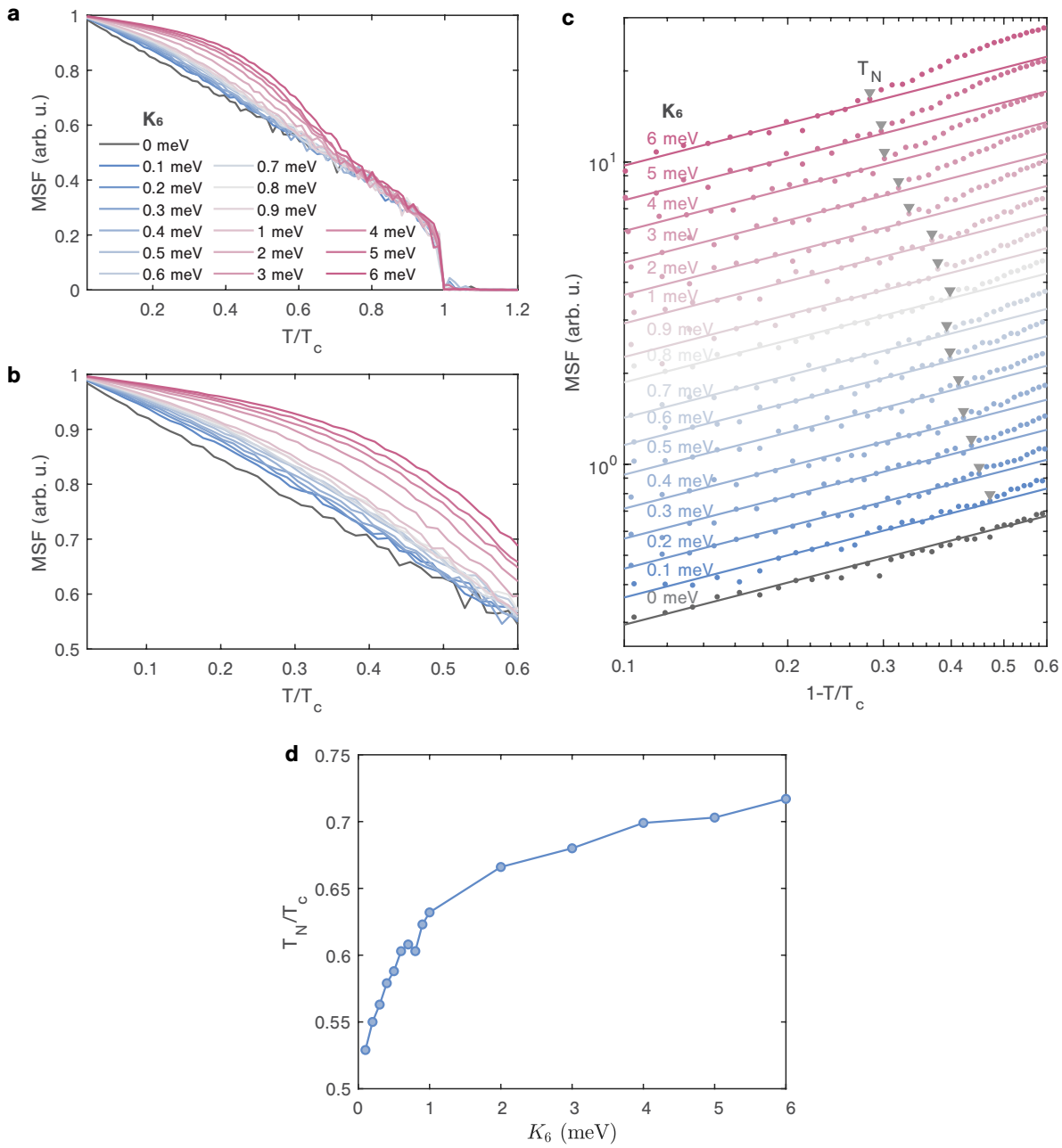


Fig. S19 | Monte Carlo simulations as a function of K_6. a-c, Temperature-dependent magnetic structure factors (MSF) for monolayer $NiPS_3$ obtained from atomistic Monte Carlo simulations (see Methods) with system size $L = 100$ and varying six-fold anisotropy K_6. These data extend the results shown in Fig. 3d of the main text. In panel (c), solid lines represent high-temperature fits to a critical scaling relation with the critical exponent fixed to $\beta = 0.231$. The temperature at which the data deviate from this trend – denoted T_N – marks the onset of the six-clock antiferromagnetic phase and is indicated by gray triangles. **d,** Transition temperatures T_N, extracted from panel (c), normalized by T_c and plotted as a function of K_6.

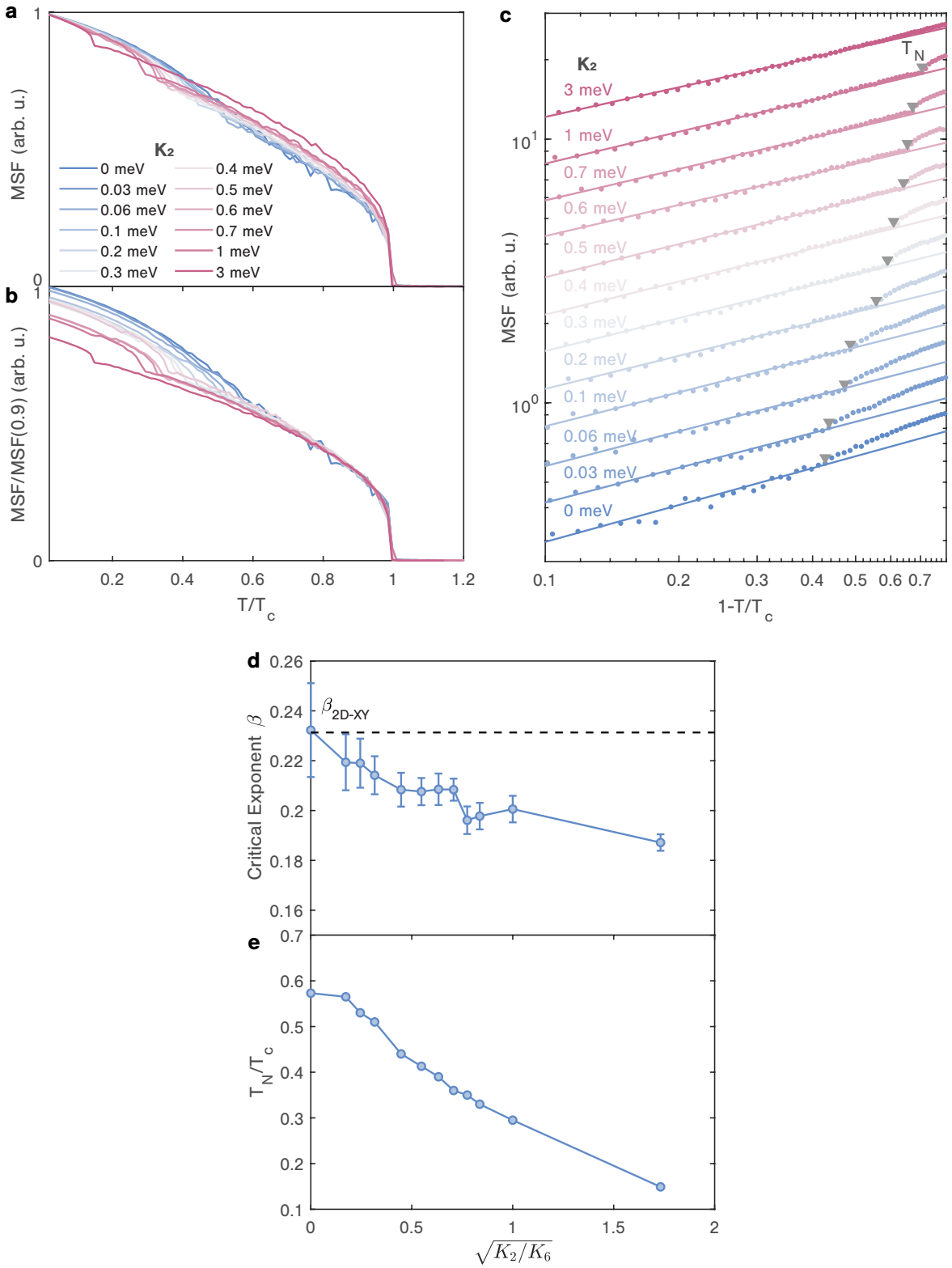


Fig. S20 | Monte Carlo simulations as a function of K_2. a-c, Temperature-dependent magnetic structure factors (MSF) for monolayer $NiPS_3$, obtained from atomistic Monte Carlo simulations (see Methods), with system size $L = 100$, fixed K_6 = 1 meV, and varying Ising anisotropy K_2. In panel (c), solid lines represent high-temperature fits to a critical scaling relation with a variable critical exponent β. The temperature at which the data deviate from this trend – denoted T_N – marks the onset of the six-clock antiferromagnetic phase and is indicated by gray triangles. **d-e,** Extracted critical exponents β and normalized transition temperatures T_N/T_c as functions of K_2, respectively. Critical exponents represent fitted values, with error bars showing 95% confidence intervals. Transition temperatures T_N are obtained from the deviation from the linear trend seen in panel (c).

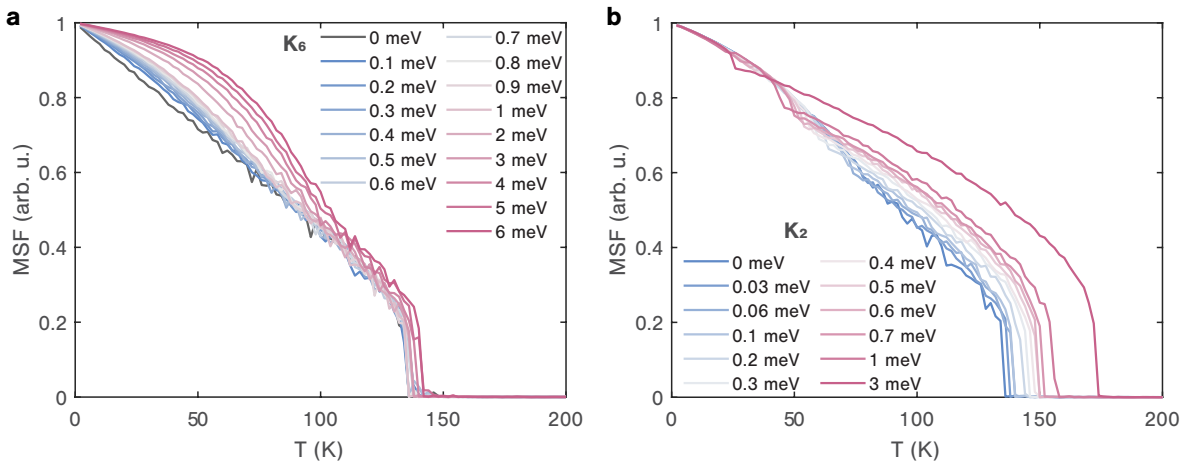


Fig. S21 | Monte Carlo simulations displayed without temperature normalization. Temperature-dependent magnetic structure factors (MSF) for monolayer $NiPS_3$, obtained from atomistic Monte Carlo simulations and plotted using the unnormalized temperature scale. The underlying data correspond to **a,** Fig. S19 and **b,** Fig. S20.

References

1. Kim, K. *et al.* Suppression of magnetic ordering in XXZ-type antiferromagnetic monolayer $NiPS_3$. *Nat. Commun.* **10**, 345 (2019).

2. Sun, Z. *et al.* Dimensionality crossover to a two-dimensional vestigial nematic state from a three-dimensional antiferromagnet in a honeycomb van der Waals magnet. *Nat. Phys.* **20**, 1764–1771 (2024).

3. Hu, L. *et al.* Observation of a magnetic phase transition in monolayer $NiPS_3$. *Phys. Rev. B* **107**, L220407 (2023).

4. Kim, T. Y. & Park, C.-H. Magnetic anisotropy and magnetic ordering of transition-metal phosphorus trisulfides. *Nano Lett.* **21**, 10114–10121 (2021).

5. Nguyen, M. H., Park, G., Park, J.-G. & Cheong, H. Interlayer interaction and Davydov splitting in antiferromagnetic few-layer $NiPS_3$. *npj Quantum Mater.* **10**, 65 (2025).

6. Skomski, R., Manchanda, P. & Kashyap, A. *Anisotropy and Crystal Field*, 103–185 (Springer International Publishing, Cham, 2021).

7. Lançon, D., Ewings, R. A., Guidi, T., Formisano, F. & Wildes, A. R. Magnetic exchange parameters and anisotropy of the quasi-two-dimensional antiferromagnet $NiPS_3$. *Phys. Rev. B* **98**, 134414 (2018).

8. José, J. V., Kadanoff, L. P., Kirkpatrick, S. & Nelson, D. R. Renormalization, vortices, and symmetry-breaking perturbations in the two-dimensional planar model. *Phys. Rev. B* **16**, 1217–1241 (1977).

9. Ueda, H. *et al.* Finite-m scaling analysis of Berezinskii-Kosterlitz-Thouless phase transitions and entanglement spectrum for the six-state clock model. *Phys. Rev. E* **101**, 062111 (2020).

10. Bramwell, S. & Holdsworth, P. Magnetization and universal sub-critical behaviour in two-dimensional XY magnets. *J. Phys. Condens. Matter* **5**, L53 (1993).

11. Fiebig, M., Pavlov, V. V. & Pisarev, R. V. Second-harmonic generation as a tool for studying electronic and magnetic structures of crystals. *J. Opt. Soc. Am. B* **22**, 96–118 (2005).

12. Sa, D., Valenti, R. & Gros, C. A generalized Ginzburg-Landau approach to second harmonic generation. *Eur. Phys. J. B* **14**, 301–305 (2000).

13. Fiebig, M. *et al.* Second harmonic generation in the centrosymmetric antiferromagnet NiO. *Phys. Rev. Lett.* **87**, 137202 (2001).

14. Chu, H. *et al.* Linear magnetoelectric phase in ultrathin $MnPS_3$ probed by optical second harmonic generation. *Phys. Rev. Lett.* **124**, 027601 (2020).

15. Kang, S. *et al.* Coherent many-body exciton in van der Waals antiferromagnet $NiPS_3$. *Nature* **583**, 785–789 (2020).

16. Birss, R. R. *Symmetry and Magnetism: Series of Monographs on Selected Topics in Solid State Physics* (North-Holland Publishing, 1966), 2nd edn.